

MÒDUL 0375 – Serveis de xarxa i Internet**Serveis de xarxa i Internet**Activitat pràctica AP1
DNS i DHCP en Windows ServerRA1. Administra serveis de resolució de noms, analitzant-los i garantint la seguretat del servei.
RA2. Administra serveis de configuració automàtica, identificant-los i verificant la correcta assignació dels paràmetres.**Observacions**

Responen en un color diferenciador.

L'objectiu d'aquest exercici és aprofundir en els conceptes i implementació dels serveis de DNS i DHCP en Windows Server.

Aquesta activitat es realitzarà en grups de 3 alumnes i es corregirà a classe en CHECK i serà part de l'avaluació del resultat d'aprenentatge de la RA1 i RA2 del mòdul de Serveis de xarxa i Internet.

Per realitzar aquesta exercici necessiteu fer servir 6 màquines virtuals que seran proporcionades en l'entorn d'ISARDVDI:

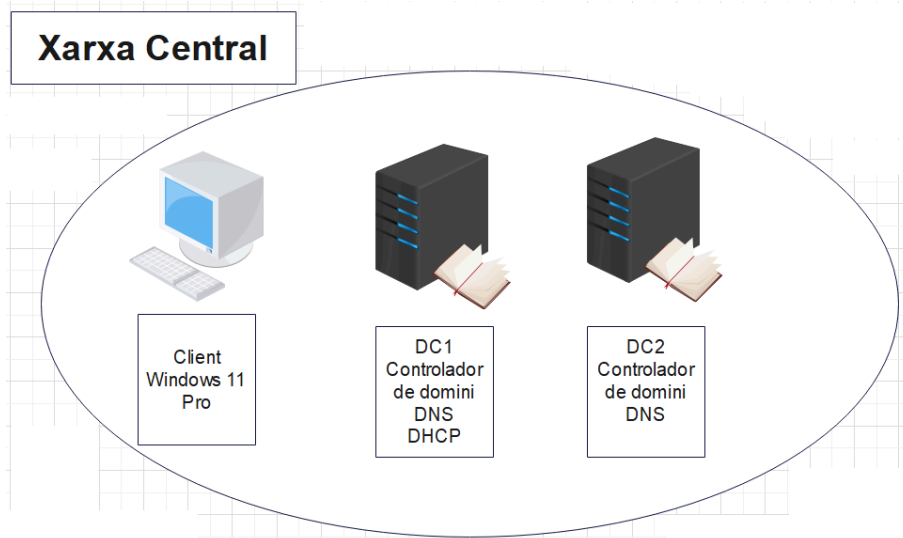
- **DC1** Controlador del domini grupX.itb.cat (Rol d'Active Directory Services, DNS i DHCP)
- **DC2** Controlador de domini grupX.itb.cat (Rol d'Active Directory Services i DNS)
- **Windows Server** amb les funcions de Router i DHCP Relay Agent amb 2 interfícies de xarxa
- **Client Windows 11 Pro** (anomenat grupXwin) ubicat a la Xarxa Central i unida al domini grupx.itb.cat
- **Client Windows 11 Pro** (anomenat grupXwremot) ubicat a la Xarxa Remota i unida al domini grupx.itb.cat
- **Client Ubuntu Desktop** (anomenat grupXuremot) ubicat a la Xarxa Remota amb connectivitat IP a la xarxa central

Xarxa central: 192.168.X.0/24**Xarxa remota : 172.16.X.0/24****Distribució equips per grup/alumne:**

Alumne 1 (Marc)	Alumne 2 (Walid)
DC1	DC2
Client Ubuntu Desktop ubicat en Xarxa Remota	Windows Server (Router i DHCP Relay agent)
Client Windows 11 Pro ubicat en Xarxa Remota	Client Windows 11 Pro ubicat en Xarxa Central

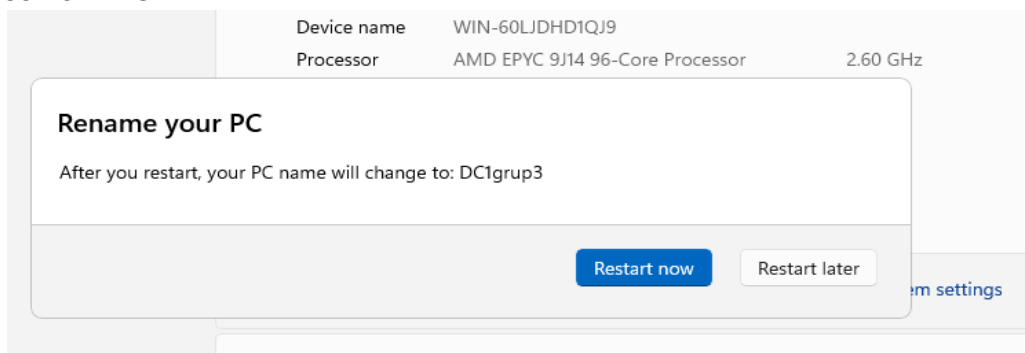
Xarxa Central (Directori Actiu, DNS's i DHCP) (4 punts)

Una empresa ubicada en Barcelona disposa de la següent infraestructura:



Volem implementar els següents serveis de xarxa en la seva xarxa central:

- Controlador de domini anomenat DC1grupx amb el servei DNS integrat a Directori Actiu i servei DHCP



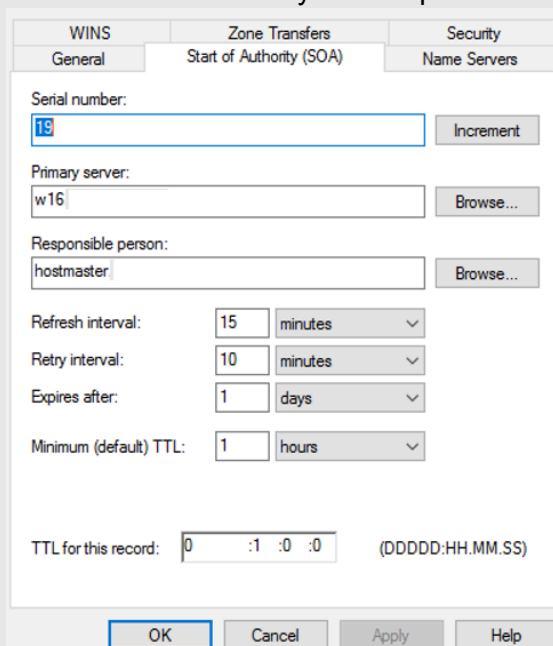
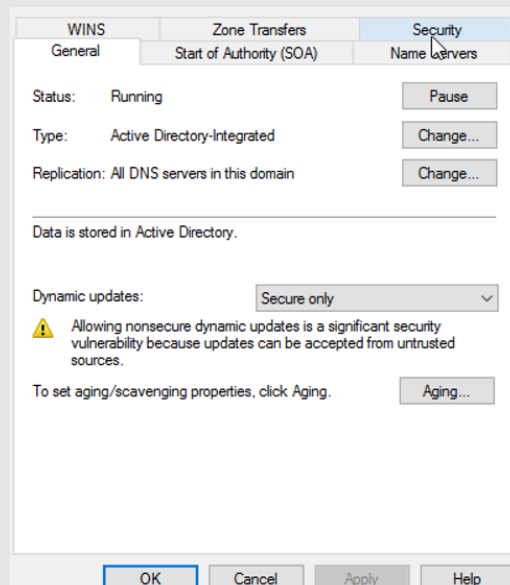
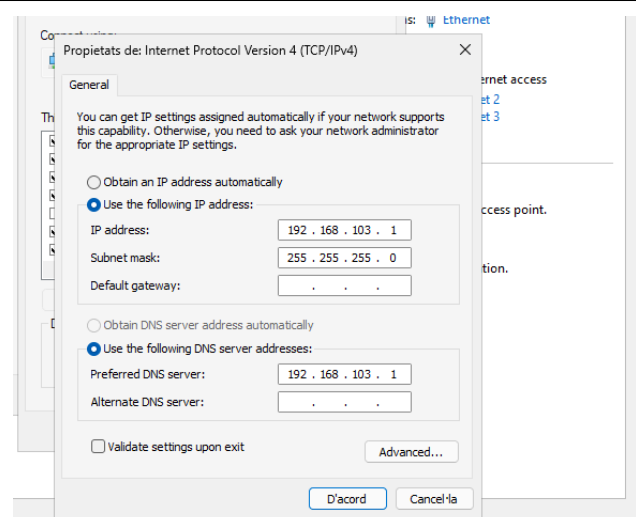
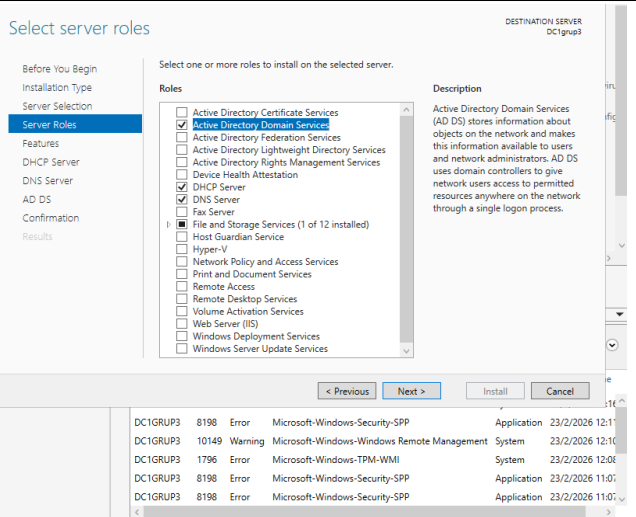
- Controlador de domini anomenat DC2grupx amb el servei DNS integrat a Directori Actiu

3922245097@edu.gencat.cat

Instal·lació i implementació del controladors de domini amb el servei DNS integrat a Directori Actiu.

- Captura de pantalla dels complements DNS en DC1 i DC2 (visualització dels registres de la zona directa del domini grupX.itb.cat i verificació replicació registres entre els DNS).
- Captura de pantalla de les propietats de la zona Inici d'autoritat (SOA) i Pestanya General amb Dynamic Updates Secure only. Configurar Aging/scavenging a 8 dies

1.

Indiqueu aquí la vostra resposta -

Captura de pantalla dels complements DNS en DC1 i DC2 (visualització dels registres de la zona directa del domini grupX.itb.cat i verificació replicació registres entre els DNS).

PROPERTIES

Add Roles and Features Wizard

Installation progress

DESTINATION SERVER
DC1grup3

Before You Begin
Installation Type
Server Selection
Server Roles
Features
DHCP Server
DNS Server
AD DS
Confirmation
Results

View installation progress

i Starting installation

Active Directory Domain Services
DHCP Server
DNS Server
Group Policy Management
Remote Server Administration Tools
Role Administration Tools
DHCP Server Tools
AD DS and AD LDS Tools
AD DS Tools
AD DS Snap-Ins and Command-Line Tools
Active Directory Administrative Center

i You can close this wizard without interrupting running tasks. View task progress or open this page again by clicking Notifications in the command bar, and then Task Details.

[Export configuration settings](#)

WELCOME TO SERVER MANAGER

QUICK START

WHAT'S NEW

LEARN MORE

ROLES AND SERVER GROUPS

Post-deployment Configuration

Configuration required for DHCP Server at DC1GRUP3
[Complete DHCP configuration](#)

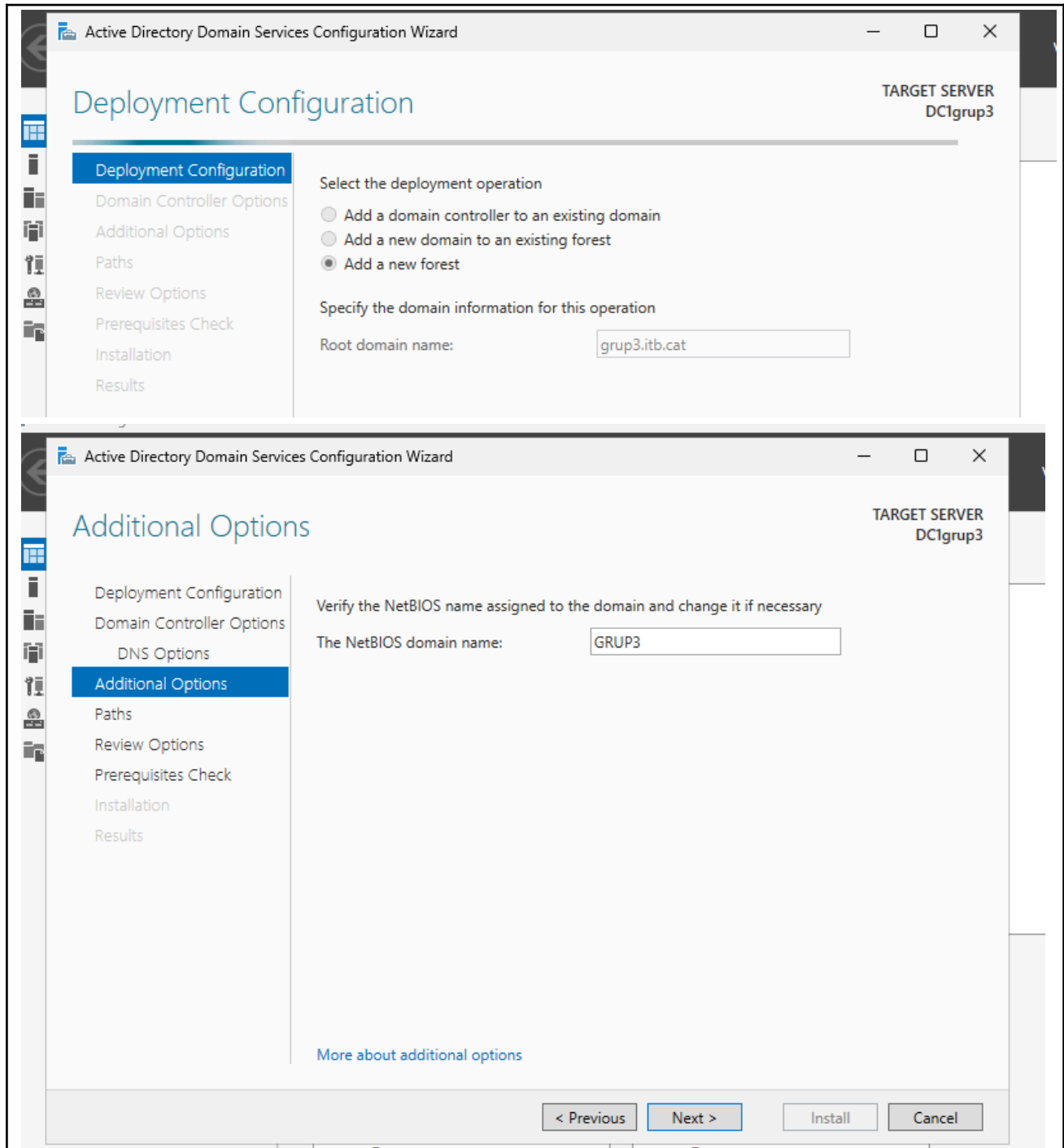
Post-deployment Configuration

Configuration required for Active Directory Domain Services at DC1GRUP3
[Promote this server to a domain controller](#)

Feature installation

Configuration required. Installation succeeded on DC1grup3.
[Add Roles and Features](#)

[Task Details](#)



Active Directory Domain Services Configuration Wizard

Deployment Configuration

TARGET SERVER
DC1grup3

Deployment Configuration

- Domain Controller Options
- Additional Options
- Paths
- Review Options
- Prerequisites Check
- Installation
- Results

Select the deployment operation

- ☐ Add a domain controller to an existing domain
- ☐ Add a new domain to an existing forest
- ☒ Add a new forest

Specify the domain information for this operation

Root domain name:

Active Directory Domain Services Configuration Wizard

Additional Options

TARGET SERVER
DC1grup3

Deployment Configuration

- Domain Controller Options
- DNS Options
- Additional Options
- Paths
- Review Options
- Prerequisites Check
- Installation
- Results

Verify the NetBIOS name assigned to the domain and change it if necessary

The NetBIOS domain name:

[More about additional options](#)

< Previous Next > Install Cancel

Active Directory Domain Services Configuration Wizard

Paths

Deployment Configuration

Domain Controller Options

DNS Options

Additional Options

Paths

Review Options

Prerequisites Check

Installation

Results

TARGET SERVER
DC1grup3

Specify the location of the AD DS database, log files, and SYSVOL

Database folder:

C:\WINDOWS\NTDS

Log files folder:

C:\WINDOWS\NTDS

SYSVOL folder:

C:\WINDOWS\SYSVOL

Results

TARGET SERVER
DC1grup3

✓ This server was successfully configured as a domain controller

Show more

✕

View detailed operation results

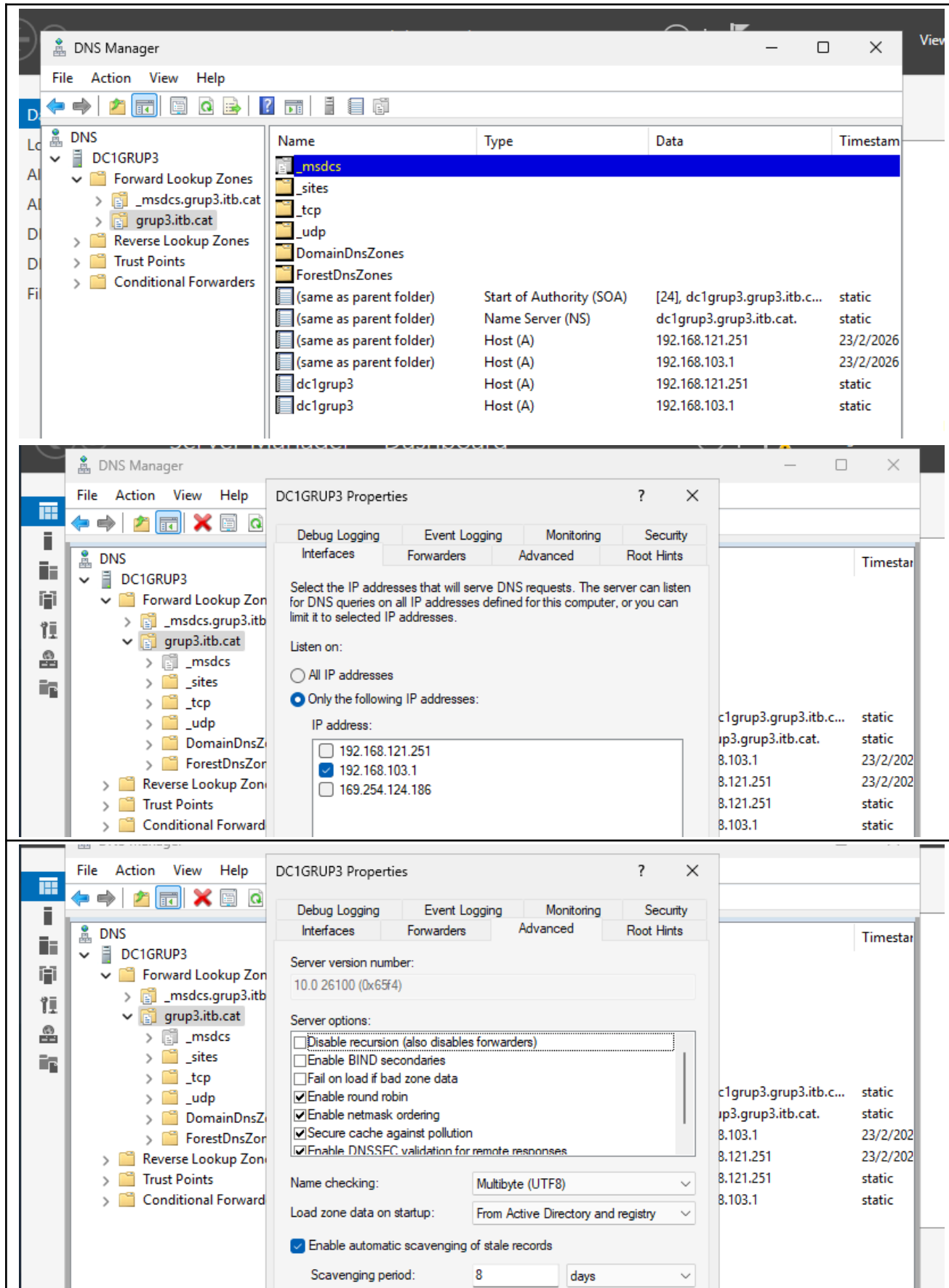
⚠ This computer has at least one physical network adapter that does not have static IP address(es) assigned to its IP Properties. If both IPv4 and IPv6 are enabled for a network adapter, both IPv4 and IPv6 static IP addresses should be assigned to both IPv4 and IPv6 Properties of the physical network adapter. Such static IP address(es) assignment should be done to all the physical network adapters for reliable Domain Name System (DNS) operation.

⚠ A delegation for this DNS server cannot be created because the authoritative parent zone cannot be found or it does not run Windows DNS server. If you are integrating with an existing DNS infrastructure, you should manually create a delegation to this DNS server in the parent zone to ensure reliable name resolution from outside the domain "grup3.itb.cat". Otherwise, no action is required.

Captura de pantalla de les propietats de la zona Inici d'autoritat (SOA) i Pestanya General

Nom del Document: AP1 - RA1-RA2 DNS i DHCP Windows Server
Elaborat: Xavier Castejón

Data: Febrer 2025
pàg. 6 de 46

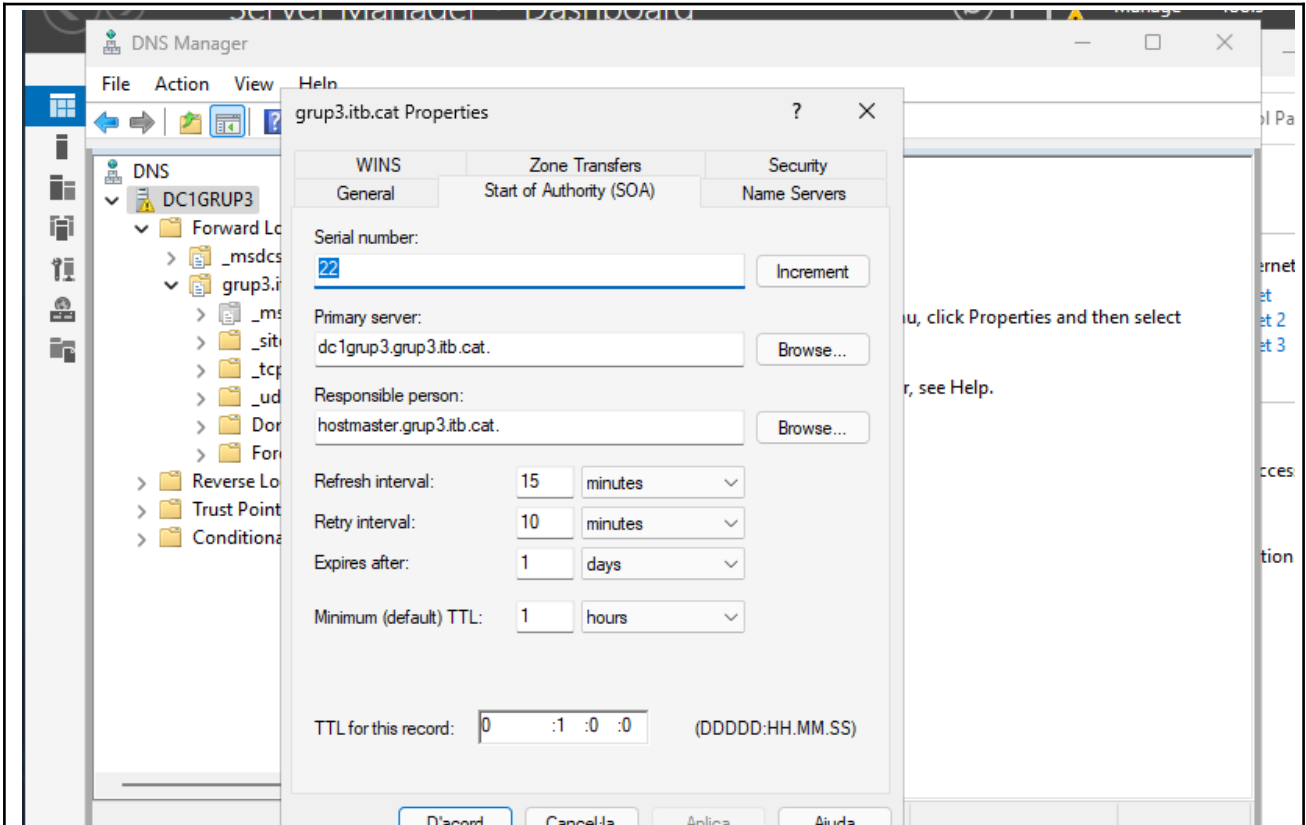


The first screenshot shows the DNS Manager console tree with DC1GRUP3 expanded, showing Forward Lookup Zones, Reverse Lookup Zones, Trust Points, and Conditional Forwarders. The main pane displays a list of DNS records for the _msdcs zone.

Name	Type	Data	Timestam
_msdcs			
_sites			
_tcp			
_udp			
DomainDnsZones			
ForestDnsZones			
(same as parent folder)	Start of Authority (SOA)	[24], dc1grup3.grup3.itb.c...	static
(same as parent folder)	Name Server (NS)	dc1grup3.grup3.itb.cat.	static
(same as parent folder)	Host (A)	192.168.121.251	23/2/2026
(same as parent folder)	Host (A)	192.168.103.1	23/2/2026
dc1grup3	Host (A)	192.168.121.251	static
dc1grup3	Host (A)	192.168.103.1	static

The second screenshot shows the DC1GRUP3 Properties dialog box, specifically the Forwarders tab. It displays the 'Listen on:' section with the option 'Only the following IP addresses:' selected. The IP address list includes 192.168.121.251, 192.168.103.1, and 169.254.124.186.

The third screenshot shows the DC1GRUP3 Properties dialog box, specifically the Advanced tab. It displays the 'Server options:' section with various checkboxes for DNS settings, including 'Enable recursion (also disables forwarders)', 'Enable BIND secondaries', 'Fail on load if bad zone data', 'Enable round robin', 'Enable netmask ordering', 'Secure cache against pollution', and 'Enable DNSSEC validation for remote responses'. The 'Name checking:' dropdown is set to 'Multibyte (UTF8)' and the 'Load zone data on startup:' dropdown is set to 'From Active Directory and registry'.



```

C:\Users\Administrator>ping grup3.itb.cat

Pinging grup3.itb.cat [192.168.103.1] with 32 bytes of data:
Reply from 192.168.103.1: bytes=32 time<1ms TTL=128
Reply from 192.168.103.1: bytes=32 time<1ms TTL=128
Reply from 192.168.103.1: bytes=32 time<1ms TTL=128
Reply from 192.168.103.1: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.103.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\Administrator>nslookup
Default Server:  UnKnown
Address:  192.168.103.1

> grup3.itb.cat
Server:  UnKnown
Address:  192.168.103.1

Name:    grup3.itb.cat
Address:  192.168.103.1

> ^C
C:\Users\Administrator>
  
```

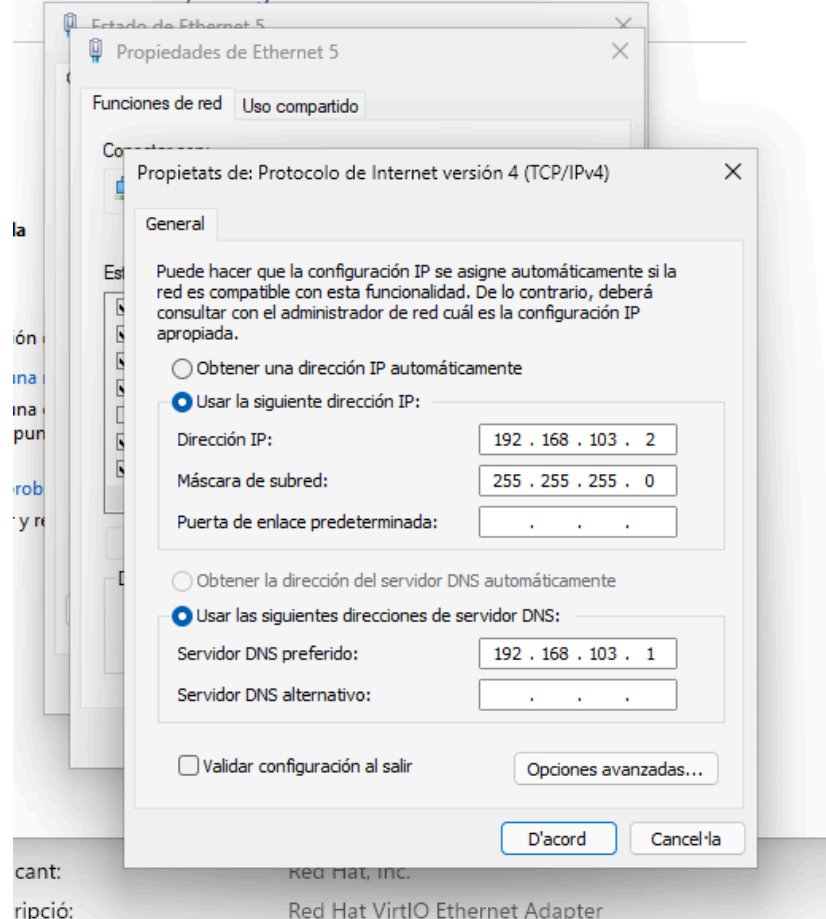

2.

- Unir equip Client Windows 11 Pro (anomenat grupXwin) ubicat a la Xarxa Central al domini grupX.itb.cat.
- Verificar que s'ha unit l'equip al domini i s'ha registrat als DNS dels controladors de domini DC1 i DC2
- Verificar en les propietats de la zona Inici d'autoritat (SOA) que el número de sèrie ha augmentat

Indiqueu aquí la vostra resposta

Verificar que s'ha unit l'equip al domini i s'ha registrat als DNS dels controladors de domini DC1 i DC2

básica de la red y configurar conexiones



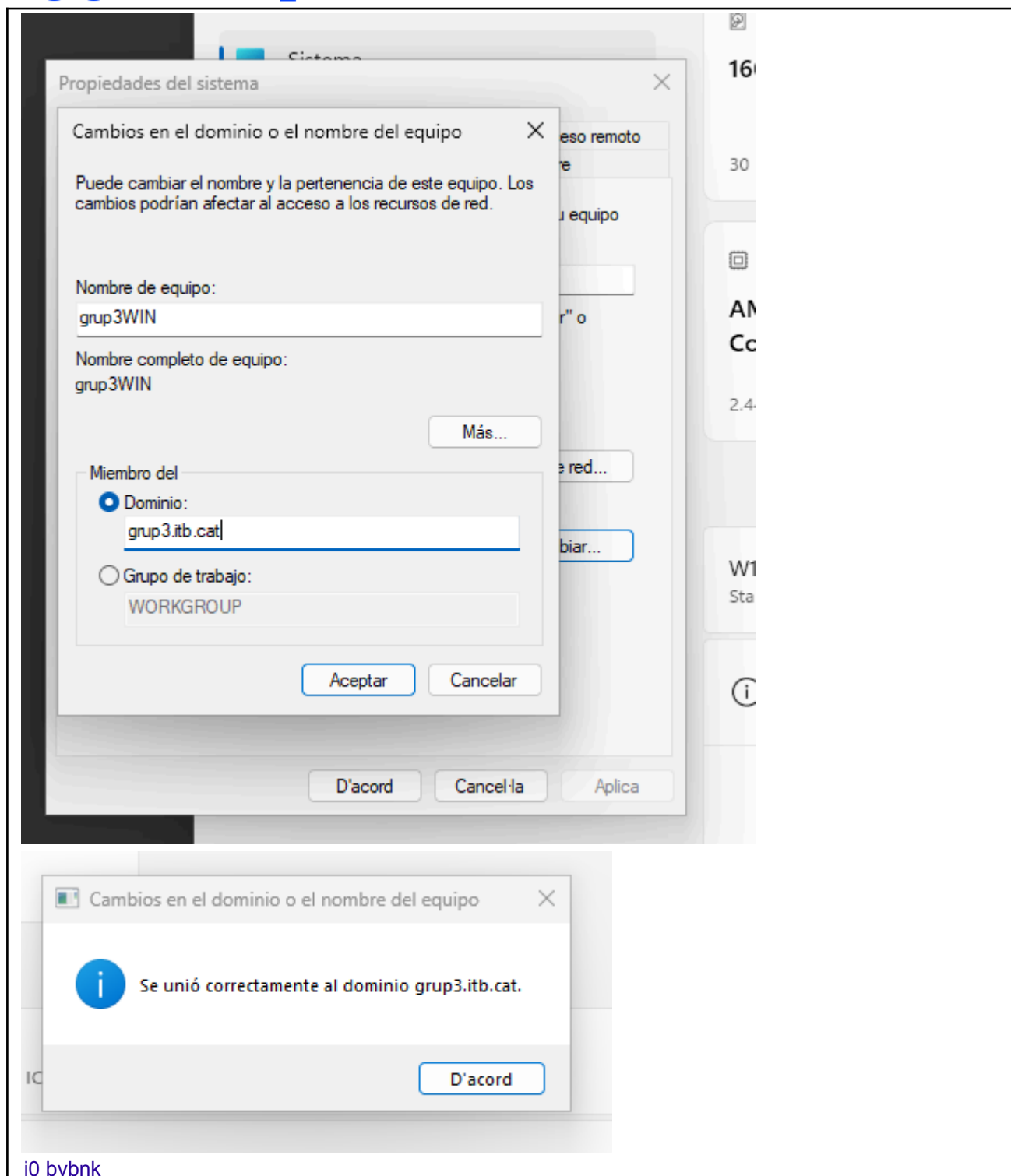
```
C:\Users\isard>ping grup3.itb.cat
```

```
Haciendo ping a grup3.itb.cat [192.168.103.1] con 32 bytes de datos:
Respuesta desde 192.168.103.1: bytes=32 tiempo=1ms TTL=128
Respuesta desde 192.168.103.1: bytes=32 tiempo=1ms TTL=128
Respuesta desde 192.168.103.1: bytes=32 tiempo=1ms TTL=128
Respuesta desde 192.168.103.1: bytes=32 tiempo=1ms TTL=128
```

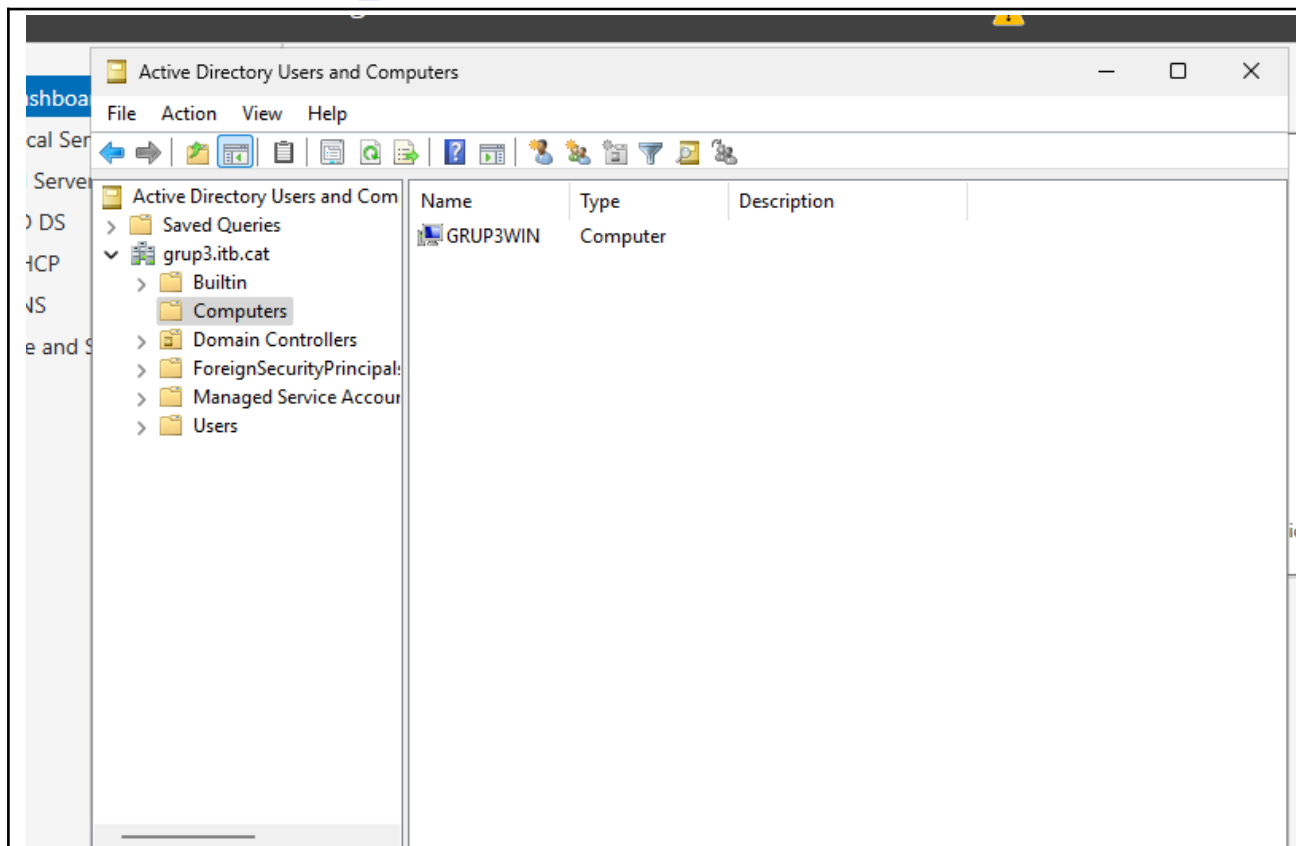
```
Estadísticas de ping para 192.168.103.1:
    Paquetes: enviados = 4, recibidos = 4, perdidos = 0
    (0% perdidos),
    Tiempos aproximados de ida y vuelta en milisegundos:
        Mínimo = 1ms, Máximo = 1ms, Media = 1ms
```

```
C:\Users\isard>
```

```
C:\Users\isard>
```



i0 bvbmk



The screenshot shows the 'Active Directory Users and Computers' window. The left pane displays the tree structure: 'Active Directory Users and Computers' > 'Saved Queries' > 'grup3.itb.cat' > 'Computers'. The right pane shows a table with the following data:

Name	Type	Description
GRUP3WIN	Computer	

Below the screenshot, there is a text instruction in purple:

Verificar en les propietats de la zona Inici d'autoritat (SOA) que el número de sèrie ha augmentat

Instal·lar i configurar el servei DHCP en el controlador de domini DC1.

Parametritzacions:

Nom de l'àmbit xarxa central: Xarxa Central grupX

Rang d'IPs: 192.168.X.100 fins 192.168.X.200.

Exclusions d'IPs: 192.168.X.1 fins 192.168.X.99

192.168.X.201 fins 192.168.X.254

El temps màxim de concessió es de 4 hores.

La porta d'enllaç es la IP del servidor que fa de router en la xarxa central

DNS Domain Name: grupX.itb.cat

Els servidors DNS seran el propi servidor DC1 i como secundari el DNS de DC2

El servidor NTP sea la IP del servidor DHCP en la xarxa.

Nom de l'àmbit xarxa remota: Xarxa Remota grupX

Rang d'IPs: 172.16.X.10 fins 192.168.X.100.

Exclusions d'IPs: 172.16.X.1 fins 172.16.X.9

172.16.X.101 fins 192.168.X.254

El temps màxim de concessió es de 8 hores.

La porta d'enllaç es la IP del servidor que fa de router en la xarxa remota

DNS Domain Name: grupX.itb.cat

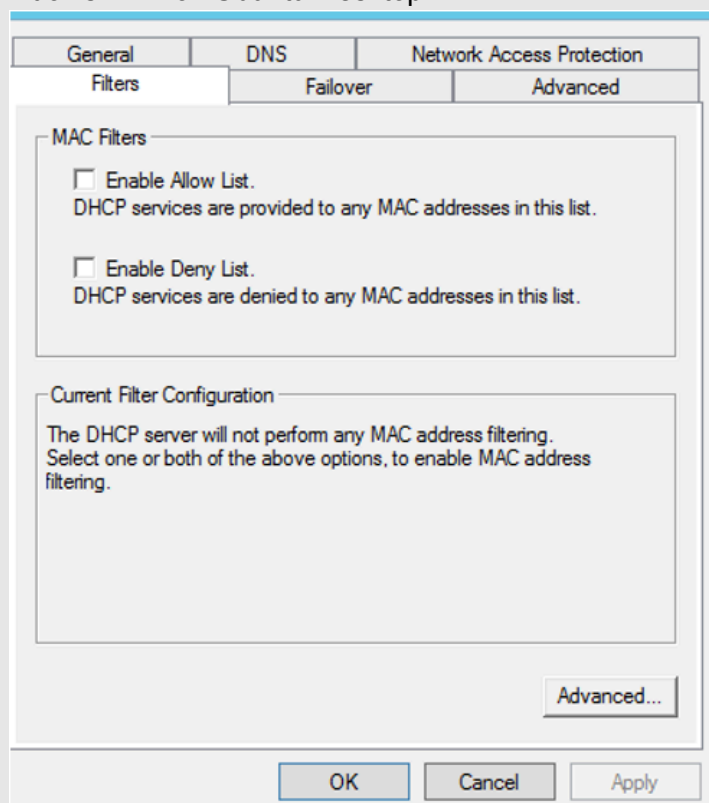
Els servidors DNS seran el propi servidor DC1 i como secundari el DNS de DC2

Tots els servidors DHCP enviaran missatges broadcast (DHCP discover) per tota la xarxa.

Realitzar un filtre per MAC address perquè només s'acceptin peticions de les vostres

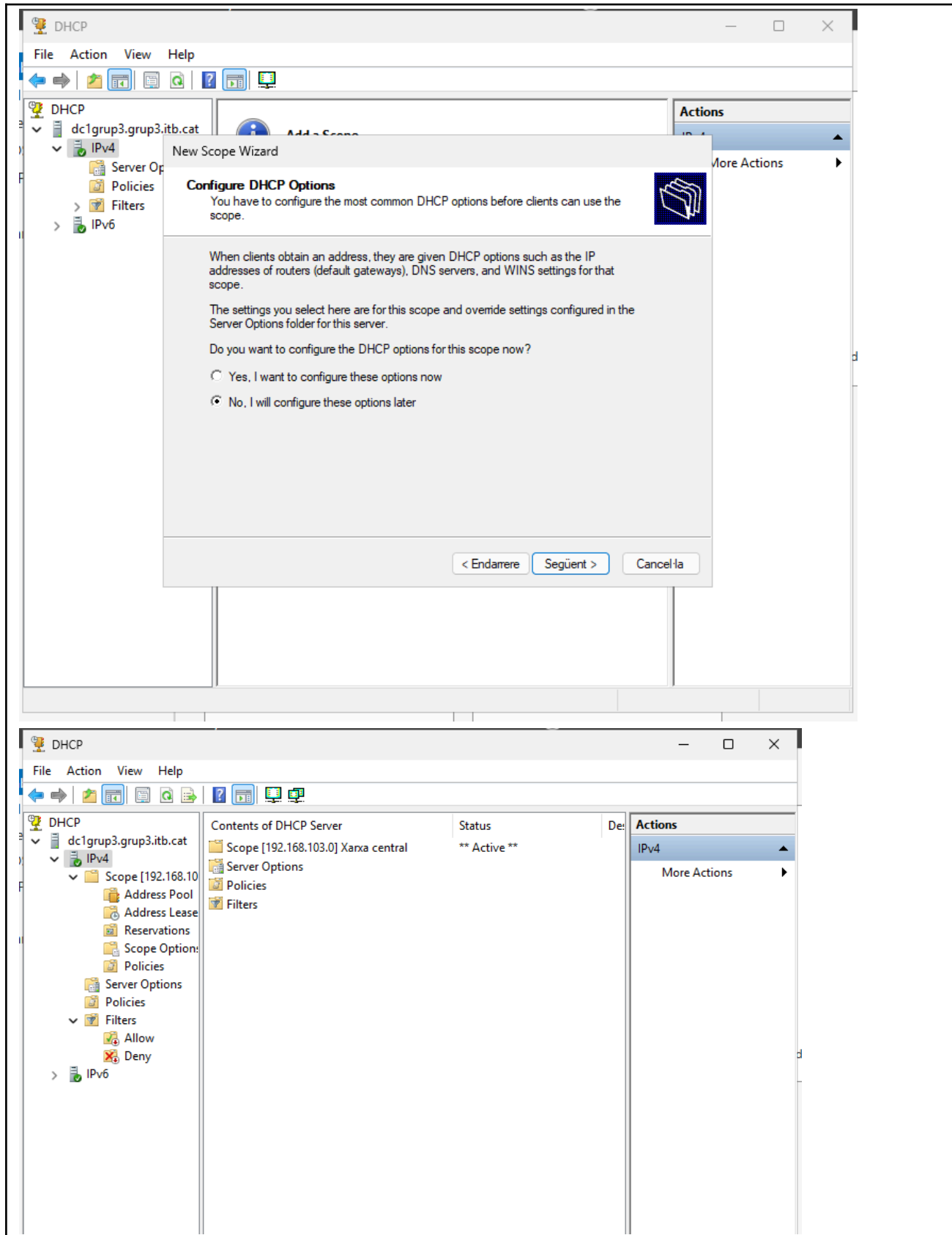
màquines client Windows 11 Pro i Ubuntu Desktop

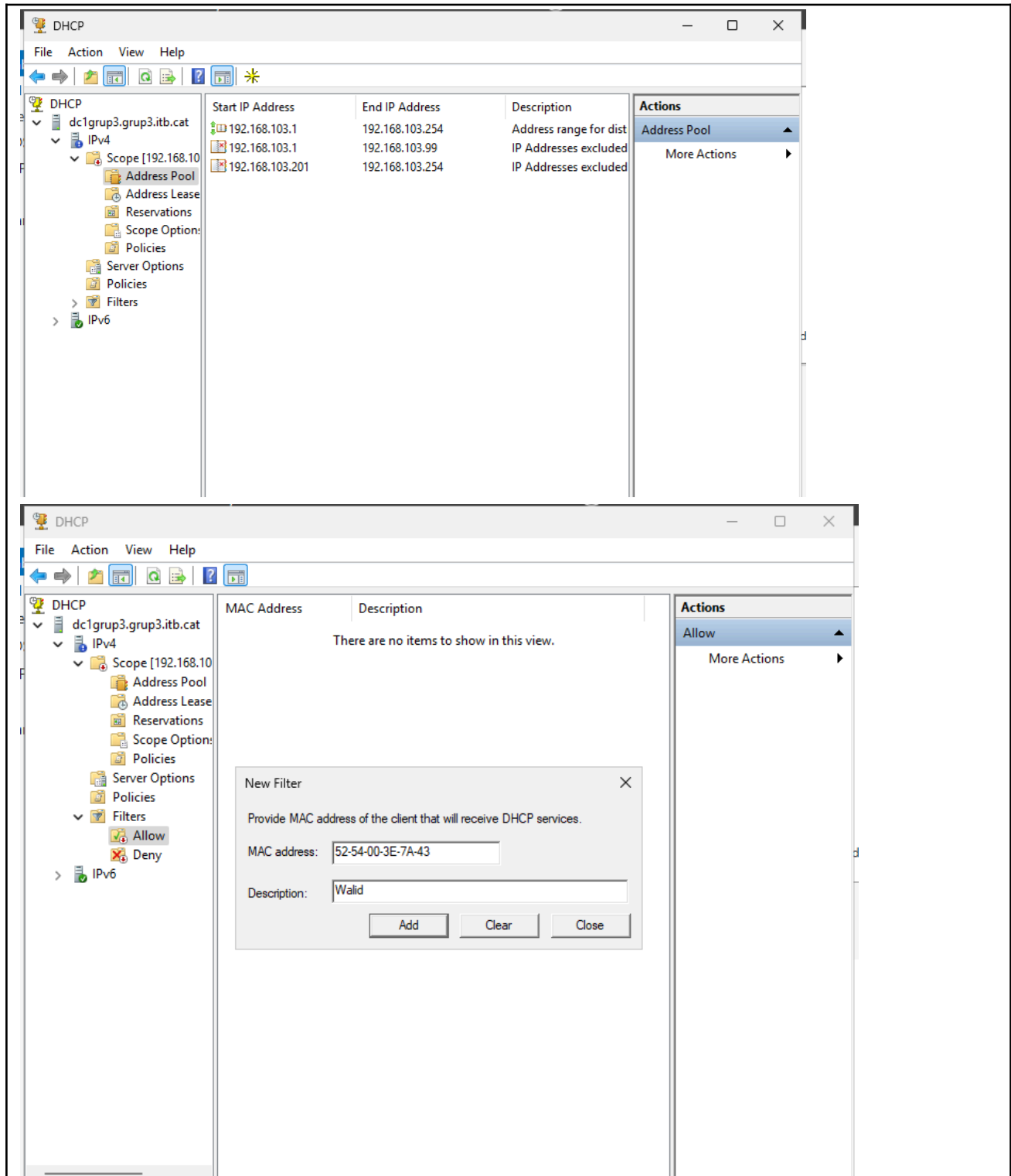
3.



The screenshot shows the 'Network Access Protection' tab in the DHCP server configuration window. The 'Filters' sub-tab is selected. Under 'MAC Filters', there are two options: 'Enable Allow List' and 'Enable Deny List', both of which are unchecked. Below these, the 'Current Filter Configuration' section states: 'The DHCP server will not perform any MAC address filtering. Select one or both of the above options, to enable MAC address filtering.' At the bottom right of the configuration area is an 'Advanced...' button. At the bottom of the window are 'OK', 'Cancel', and 'Apply' buttons.

Indiqueu aquí la vostra resposta





The image shows two screenshots of the Windows DHCP console. The top screenshot displays the 'Address Pool' tab for a DHCP scope, showing a list of IP addresses and their descriptions. The bottom screenshot shows the 'Filters' tab, where a 'New Filter' dialog box is open, allowing the user to add a new filter based on MAC address.

Top Screenshot: DHCP Console - Address Pool

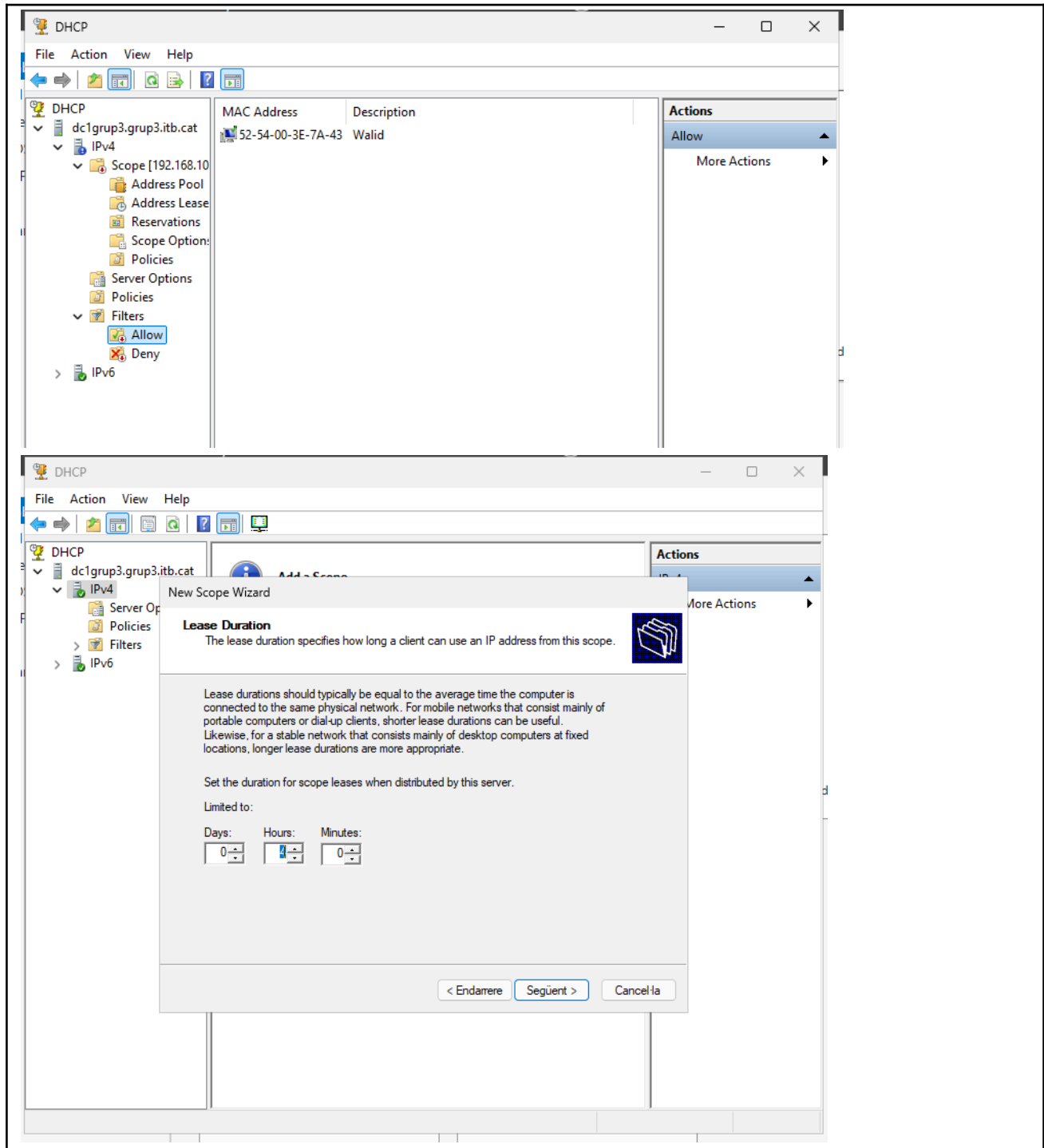
Start IP Address	End IP Address	Description
192.168.103.1	192.168.103.254	Address range for dist
192.168.103.1	192.168.103.99	IP Addresses excluded
192.168.103.201	192.168.103.254	IP Addresses excluded

Bottom Screenshot: DHCP Console - Filters

The 'New Filter' dialog box is open, showing the following fields:

- Provide MAC address of the client that will receive DHCP services.
- MAC address: 52-54-00-3E-7A-43
- Description: Valid

Buttons: Add, Clear, Close



Rename your PC

After you restart, your PC name will change to: DC2grup3

Restart now

Restart later

Propietats de: Internet Protocol Version 4 (TCP/IPv4) X

General

You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.

☐ Obtain an IP address automatically

☒ Use the following IP address:

IP address:

Subnet mask:

Default gateway:

☐ Obtain DNS server address automatically

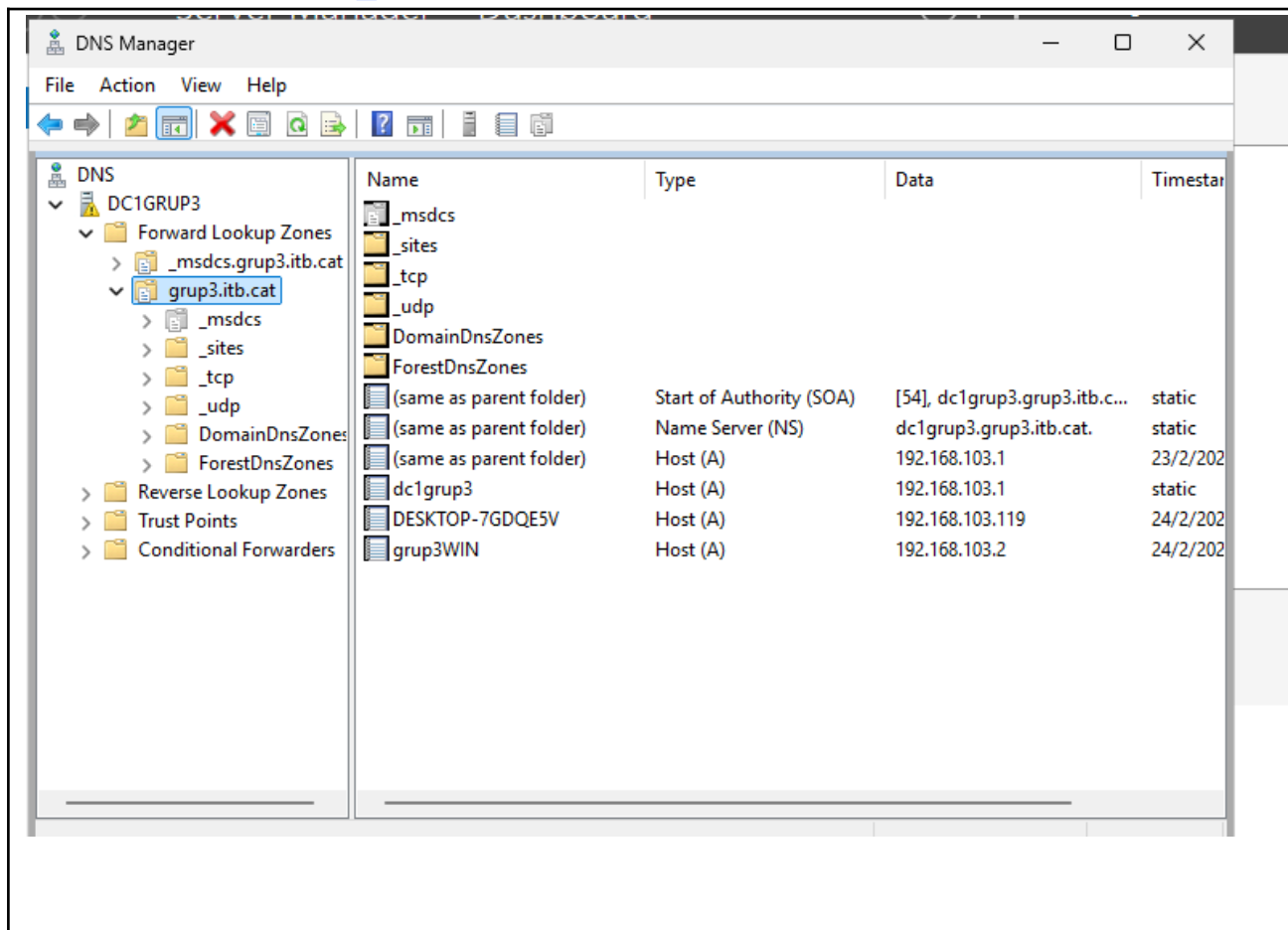
☒ Use the following DNS server addresses:

Preferred DNS server:

Alternate DNS server:

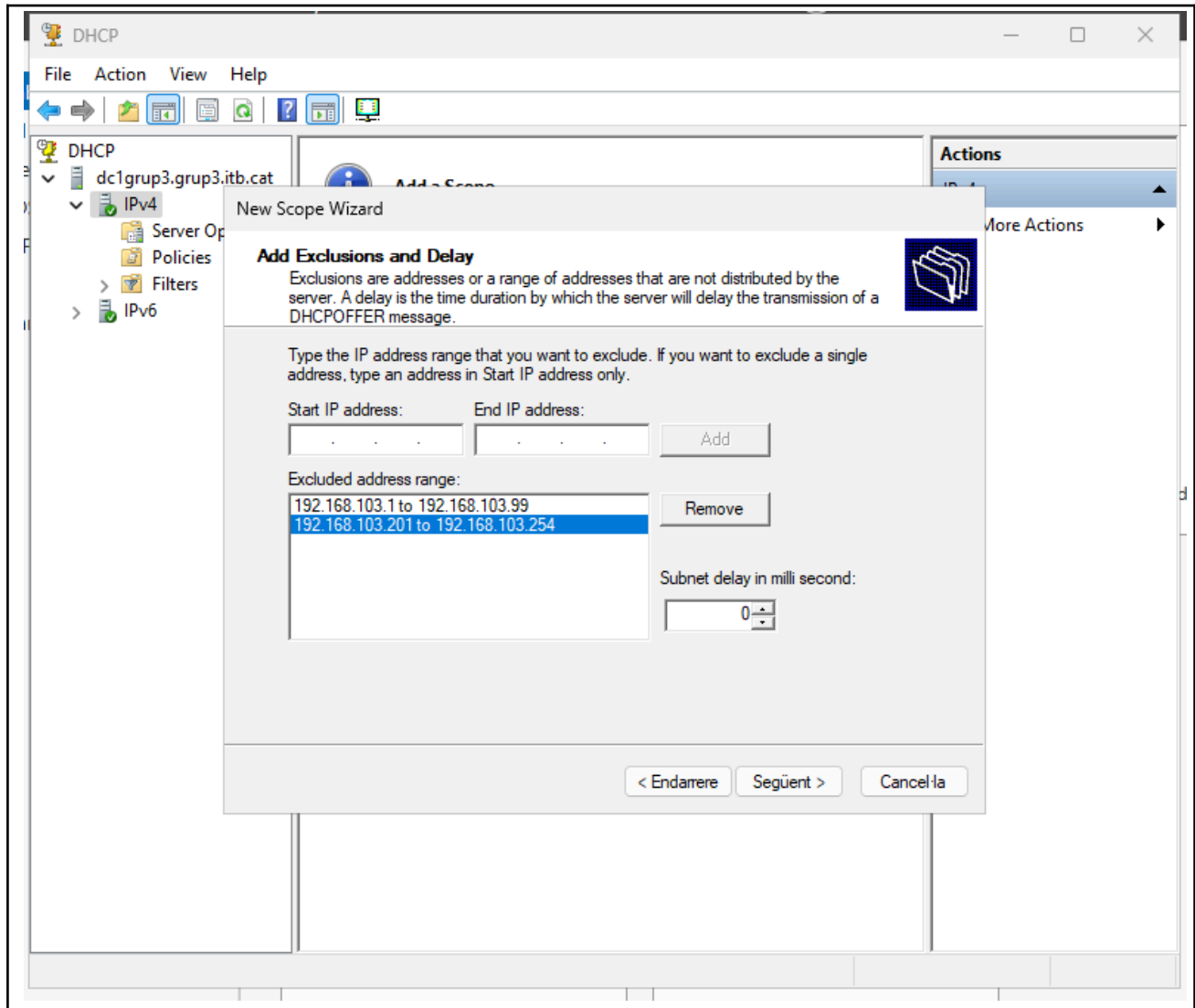
☐ Validate settings upon exit Advanced...

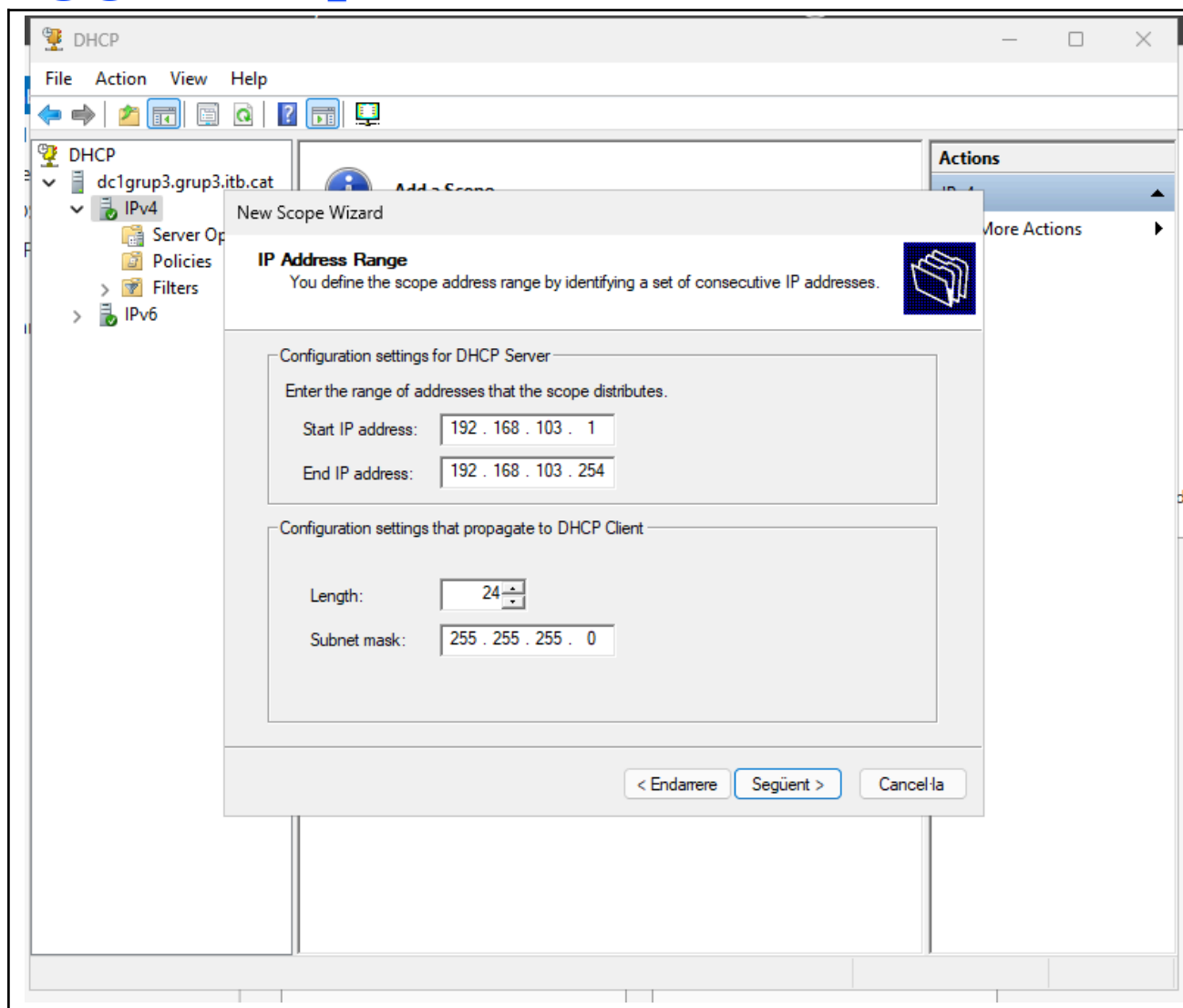
D'acord Cancel·la

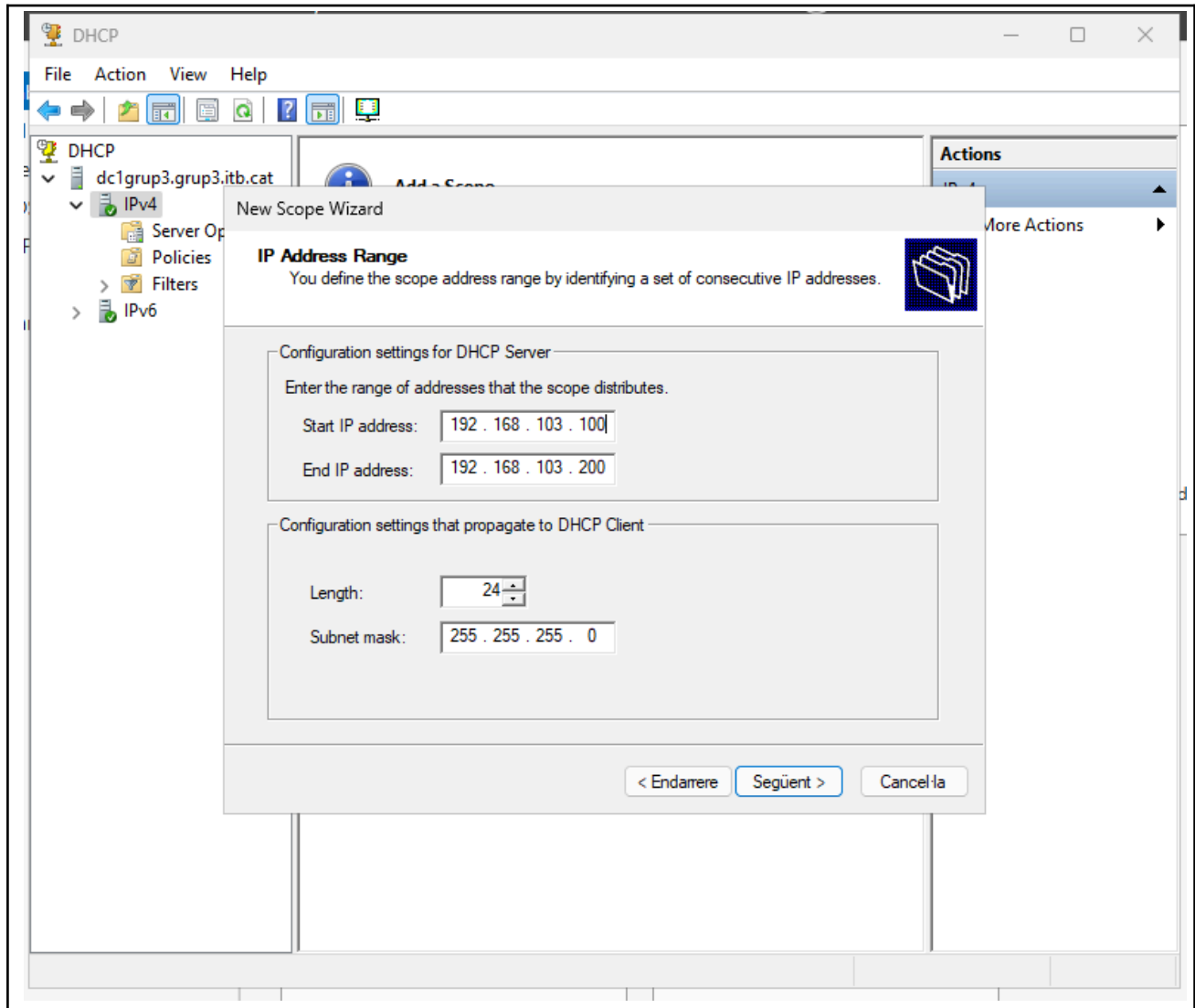


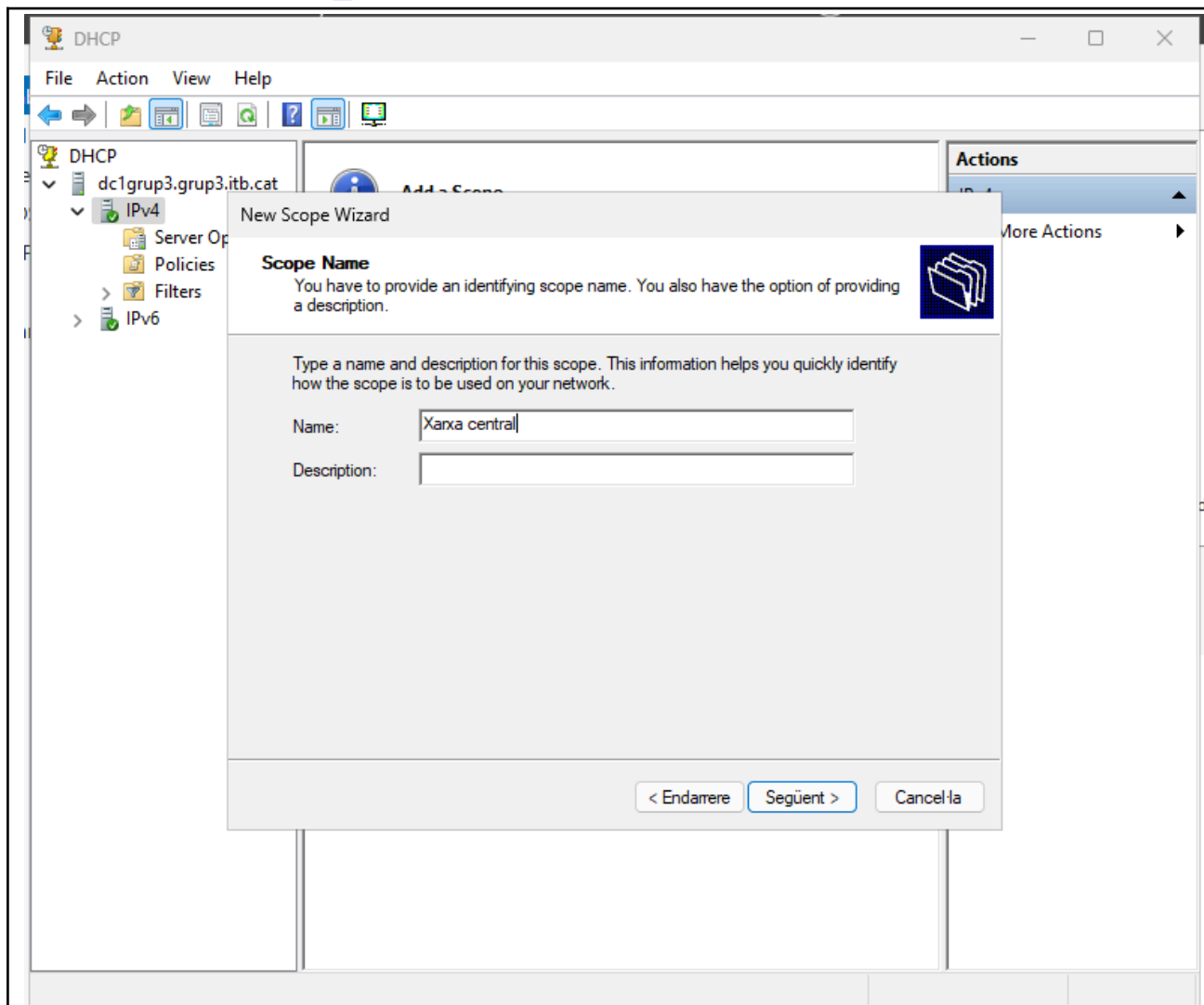
The screenshot shows the Windows DNS Manager console. The left pane displays the hierarchy: DNS > DC1GRUP3 > Forward Lookup Zones > _msdcs.grup3.itb.cat > grup3.itb.cat. The right pane shows a list of DNS records for the grup3.itb.cat zone.

Name	Type	Data	Timestamp
_msdcs			
_sites			
_tcp			
_udp			
DomainDnsZones			
ForestDnsZones			
(same as parent folder)	Start of Authority (SOA)	[54], dc1grup3.grup3.itb.c...	static
(same as parent folder)	Name Server (NS)	dc1grup3.grup3.itb.cat.	static
(same as parent folder)	Host (A)	192.168.103.1	23/2/202
dc1grup3	Host (A)	192.168.103.1	static
DESKTOP-7GDQE5V	Host (A)	192.168.103.119	24/2/202
grup3WIN	Host (A)	192.168.103.2	24/2/202









New Scope Wizard

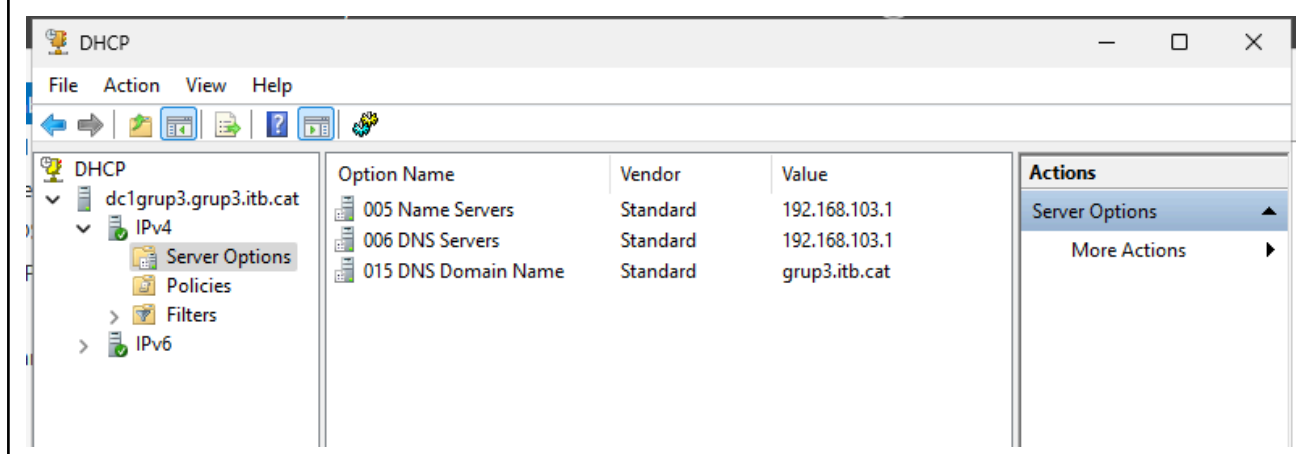
Scope Name
You have to provide an identifying scope name. You also have the option of providing a description.

Type a name and description for this scope. This information helps you quickly identify how the scope is to be used on your network.

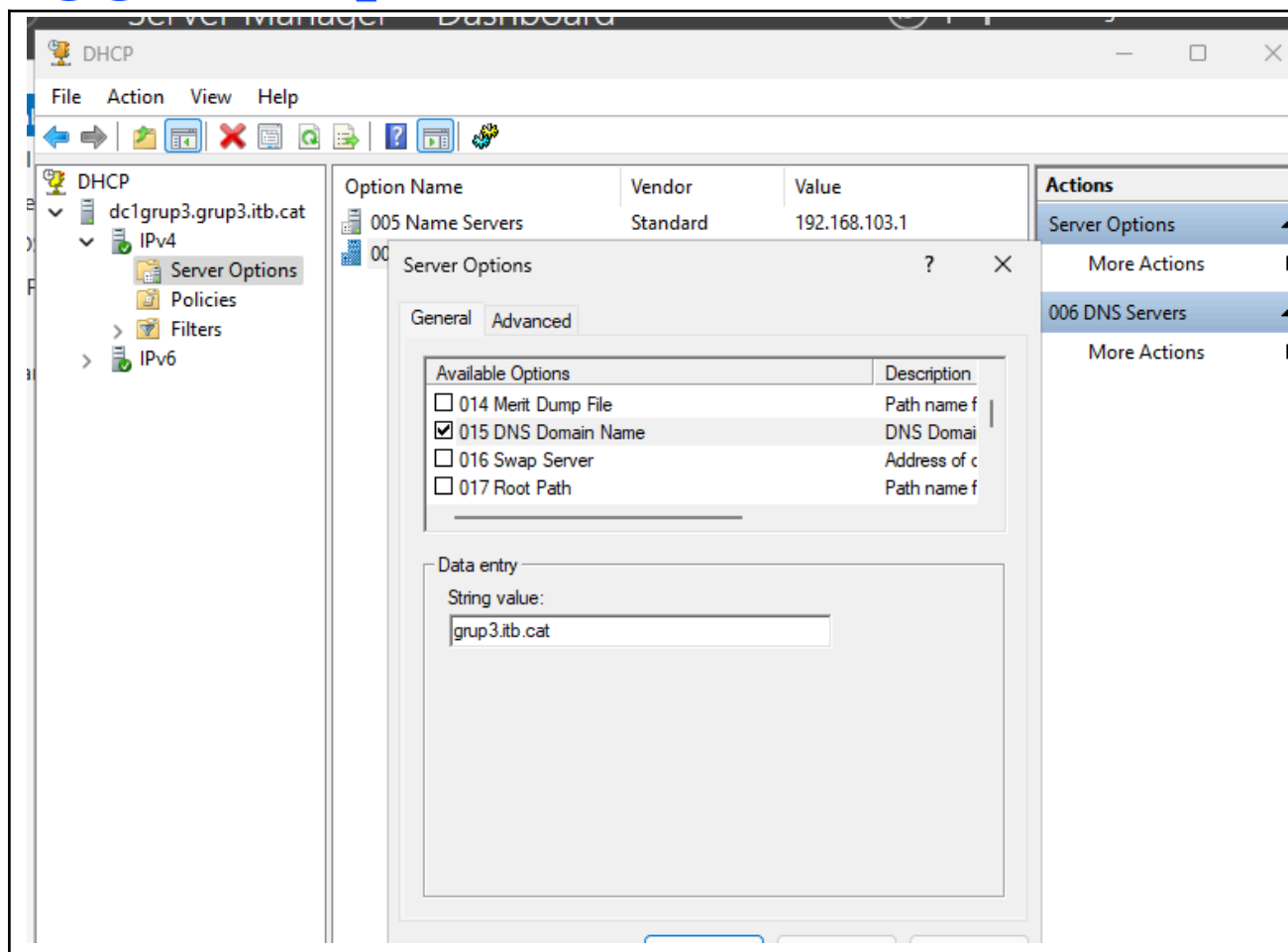
Name:

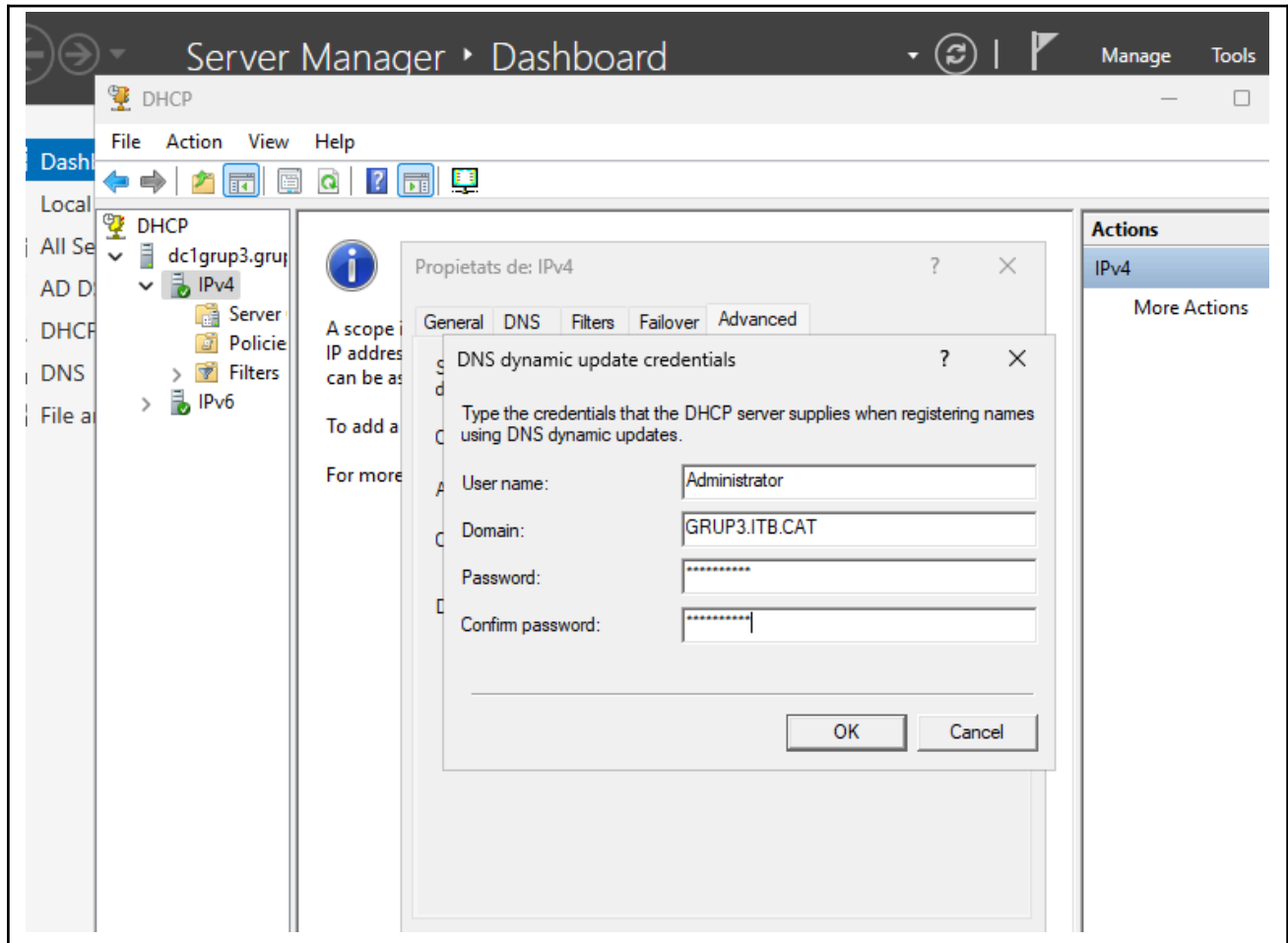
Description:

< Endarere **Següent >** Cancel·la

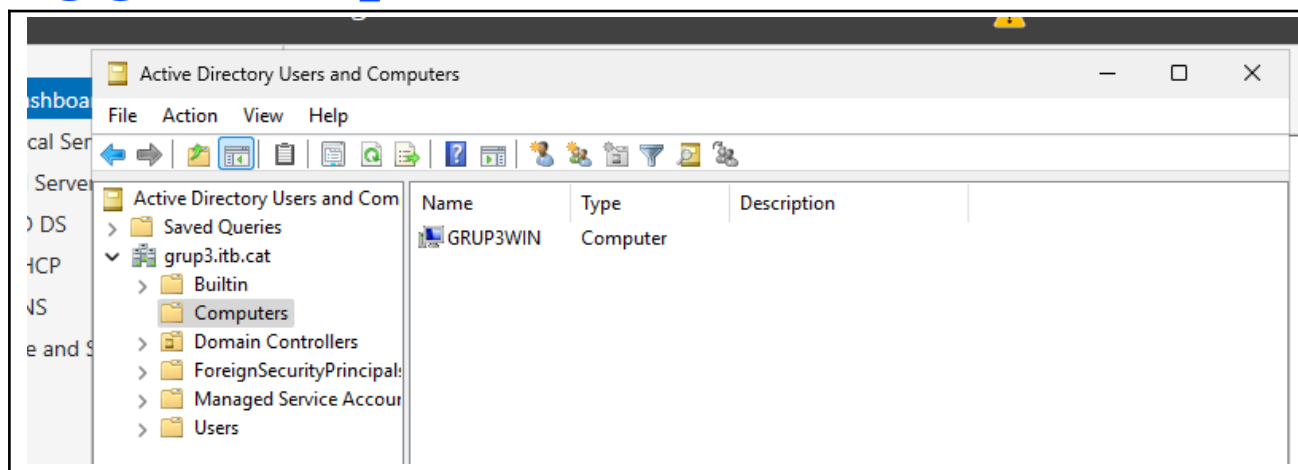


Option Name	Vendor	Value
005 Name Servers	Standard	192.168.103.1
006 DNS Servers	Standard	192.168.103.1
015 DNS Domain Name	Standard	grup3.itb.cat





[illegible]



Active Directory Domain Services Configuration Wizard

Deployment Configuration

TARGET SERVER
DC2grup3

- Deployment Configuration**
- Domain Controller Options
- Additional Options
- Paths
- Review Options
- Prerequisites Check
- Installation
- Results

Select the deployment operation

☒ Add a domain controller to an existing domain
☐ Add a new domain to an existing forest
☐ Add a new forest

Specify the domain information for this operation

Domain:

Supply the credentials to perform this operation

<No credentials provided>

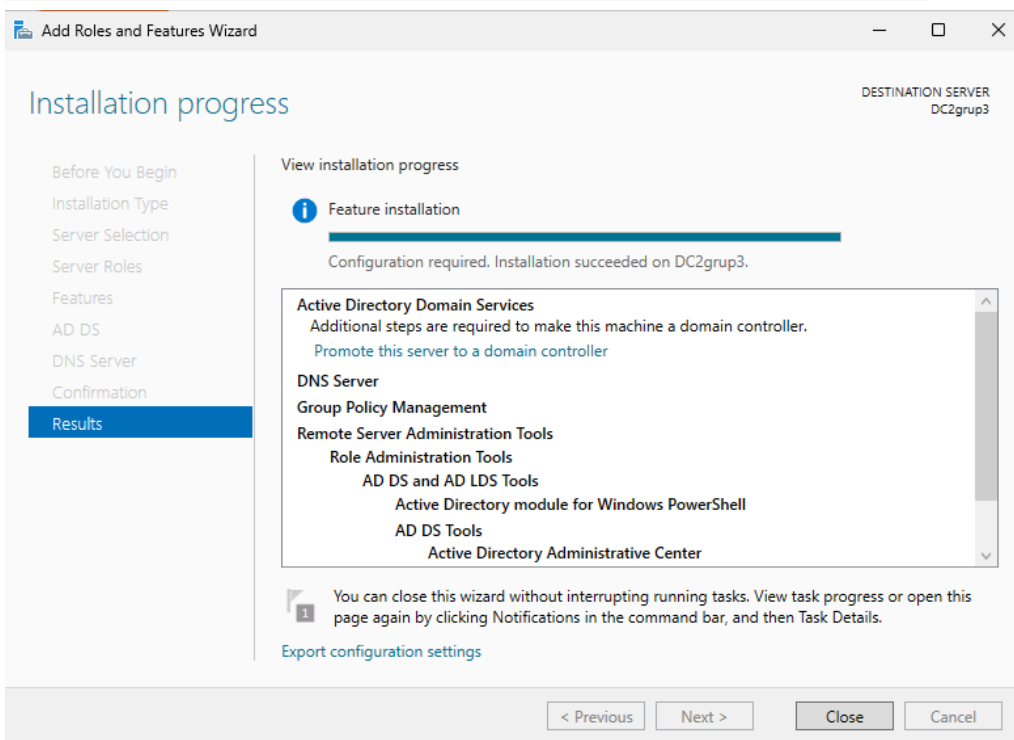
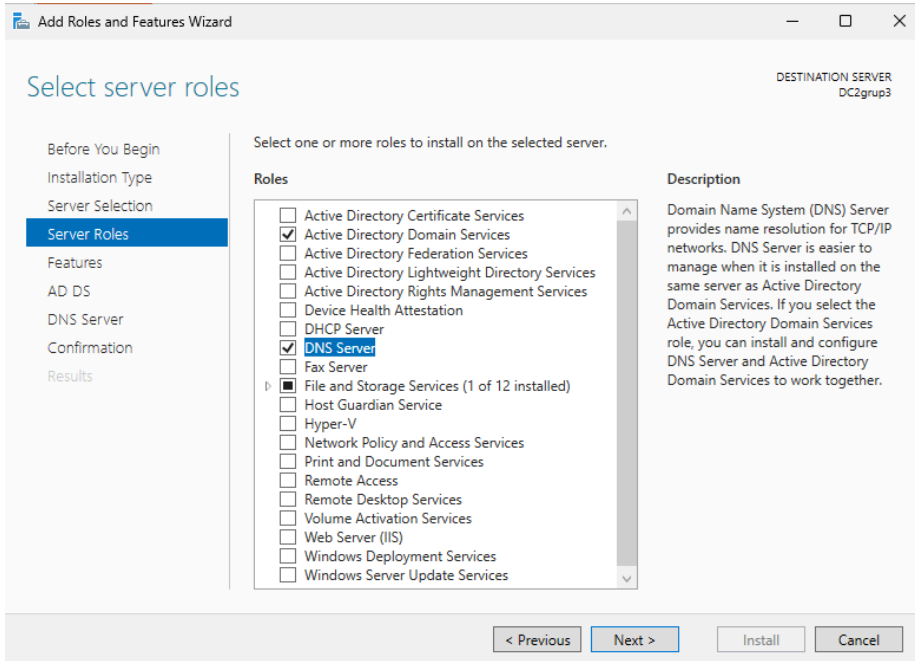
DHCP			
File Action View Help			
Local All Servers AD D DHCP DNS File a	DHCP		
	dc1grup3.grup3.itb.cat		
	IPv4		
	Server Options		
	Scope [192.168.10.0/24]		
	Address Pool		
	Address Lease		
	Reservations		
	Scope Options		
	Policies		
	Filters		
	Allow		
	Deny		
	IPv6		

Client IP Address	Name	Lease Expiration	Actions
192.168.103.100	grup6win.grup6.itb....	24/2/2026 13:57:46	Address Leases
192.168.103.101	grup5win.grup5.itb....	24/2/2026 13:59:51	More Actions
192.168.103.102	grup3WIN.grup3.itb....	24/2/2026 14:27:20	192.168.103.102
192.168.103.105	grup7win.grup7.itb....	24/2/2026 13:53:00	More Actions
192.168.103.107	DESKTOP-7GDQE5V...	24/2/2026 13:53:31	
192.168.103.108	DESKTOP-7GDQE5V...	24/2/2026 13:54:49	
192.168.103.109	DESKTOP-7GDQE5V...	24/2/2026 13:59:27	
192.168.103.110	DESKTOP-7GDQE5V...	24/2/2026 13:59:27	
192.168.103.115	DESKTOP-7GDQE5V...	24/2/2026 13:59:47	
192.168.103.116	DESKTOP-7GDQE5V...	24/2/2026 13:59:53	
192.168.103.117	DESKTOP-7GDQE5V...	24/2/2026 13:59:53	
192.168.103.118	DESKTOP-7GDQE5V...	24/2/2026 13:59:54	
192.168.103.119	DESKTOP-7GDQE5V...	24/2/2026 14:00:04	

4.

Fer que el client Windows 11 Pro unit al domini de la xarxa central agafi la configuració del DHCP.

Verificació i comprovació de la configuració automàtica pel DHCP



Active Directory Domain Services Configuration Wizard

Deployment Configuration

Deployment Configuration

Domain Controller Options

DNS Options

Additional Options

Paths

Review Options

Prerequisites Check

Installation

Results

TARGET SERVER
DC2grup3.grup3.itb.cat

Select the deployment operation

☒ Add a domain controller to an existing domain
 ☐ Add a new domain to an existing forest
 ☐ Add a new forest

Specify the domain information for this operation

Domain:

grup3.itb.cat

Select...

Supply the credentials to perform this operation

GRUP3\Administrator

Change...

Active Directory Domain Services Configuration Wizard

Domain Controller Options

Deployment Configuration

Domain Controller Options

DNS Options

Additional Options

Paths

Review Options

Prerequisites Check

Installation

Results

TARGET SERVER
DC2grup3.grup3.itb.cat

Specify domain controller capabilities and site information

☒ Domain Name System (DNS) server
 ☒ Global Catalog (GC)
 ☐ Read only domain controller (RODC)

Site name:

Default-First-Site-Name

Type the Directory Services Restore Mode (DSRM) password

Password:

Confirm password:

Active Directory Domain Services Configuration Wizard

Installation

Deployment Configuration

Domain Controller Options

DNS Options

Additional Options

Paths

Review Options

Prerequisites Check

Installation

Results

TARGET SERVER
DC2grup3.grup3.itb.cat

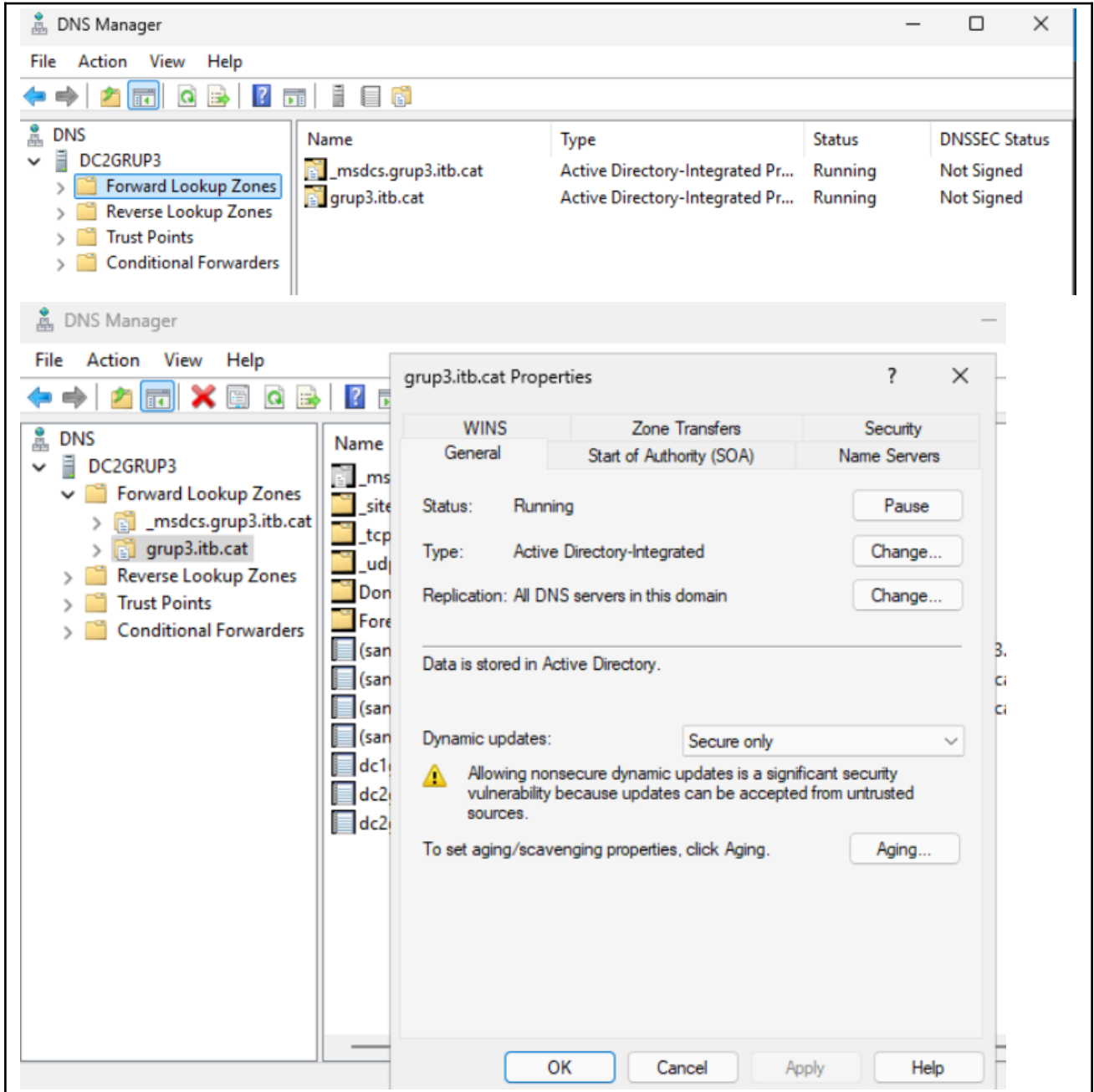
Progress

Checking domain upgrade status

View detailed operation results

Nom del Document: AP1 - RA1-RA2 DNS i DHCP Windows Server
Elaborat: Xavier Castejón

Data: Febrer 2025
pàg. 28 de 46



The image shows two overlapping screenshots of the Windows DNS Manager console. The top screenshot displays a list of DNS zones under the DC2GRUP3 server. The bottom screenshot shows the 'grup3.itb.cat Properties' dialog box, specifically the 'General' tab.

Top Screenshot: DNS Manager Console

Name	Type	Status	DNSSEC Status
_msdcs.grup3.itb.cat	Active Directory-Integrated Pr...	Running	Not Signed
grup3.itb.cat	Active Directory-Integrated Pr...	Running	Not Signed

Bottom Screenshot: grup3.itb.cat Properties - General Tab

WINS: General

Zone Transfers: Start of Authority (SOA)

Security: Name Servers

Status: Running [Pause]

Type: Active Directory-Integrated [Change...]

Replication: All DNS servers in this domain [Change...]

Data is stored in Active Directory.

Dynamic updates: Secure only (dropdown menu)

⚠️ Allowing nonsecure dynamic updates is a significant security vulnerability because updates can be accepted from untrusted sources.

To set aging/scavenging properties, click Aging... [Aging...]

[OK] [Cancel] [Apply] [Help]

grup3.itb.cat Properties

Zone Aging/Scavenging Properties

☒ Scavenge stale resource records

No-refresh interval

The time between the most recent refresh of a record timestamp and the moment when the timestamp may be refreshed again.

No-refresh interval: 8 days

Refresh interval

The time between the earliest moment when a record timestamp can be refreshed and the earliest moment when the record can be scavenged. The refresh interval must be longer than the maximum record refresh period.

Refresh interval: 8 days

OK Cancel

OK Cancel Apply Help

grup3.itb.cat Properties

WINS

Zone Transfers

Security

General

Start of Authority (SOA)

Name Servers

Serial number: 82 Increment

Primary server: dc2grup3.grup3.itb.cat. Browse...

Responsible person: hostmaster.grup3.itb.cat. Browse...

Refresh interval: 15 minutes

Retry interval: 10 minutes

Expires after: 1 days

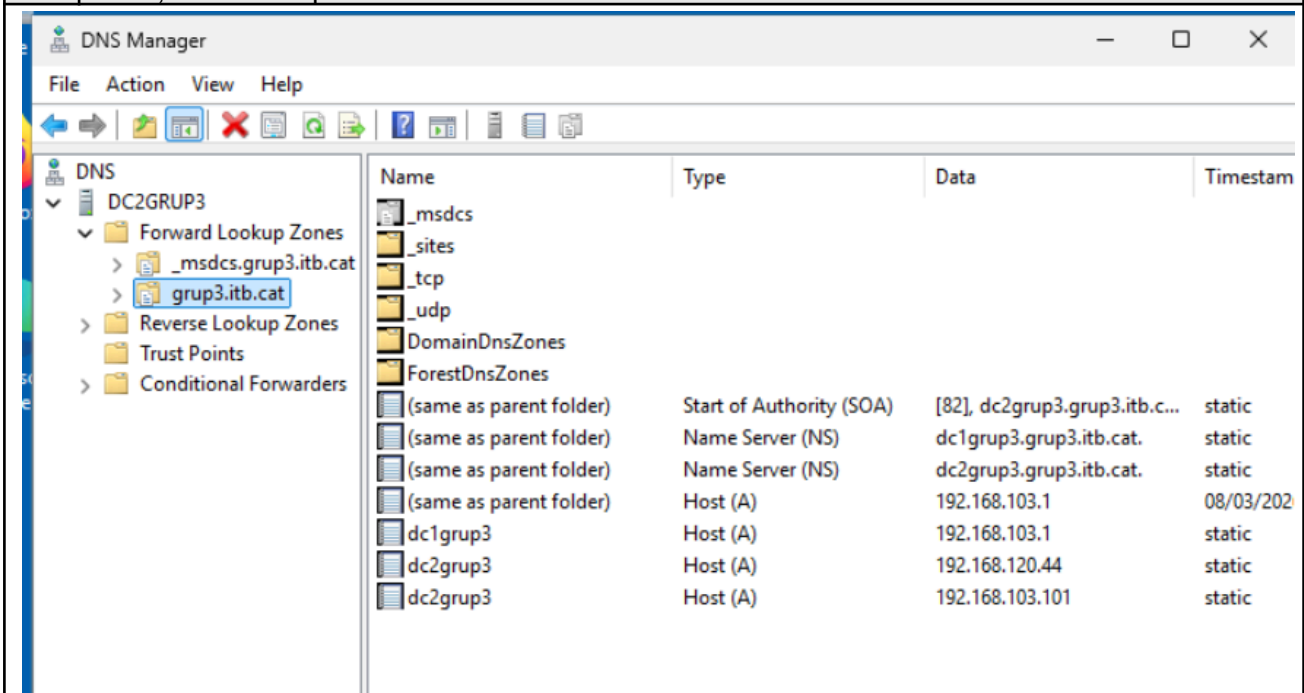
Minimum (default) TTL: 1 hours

TTL for this record: 0 :1 :0 :0 (DDDD:HH.MM.SS)

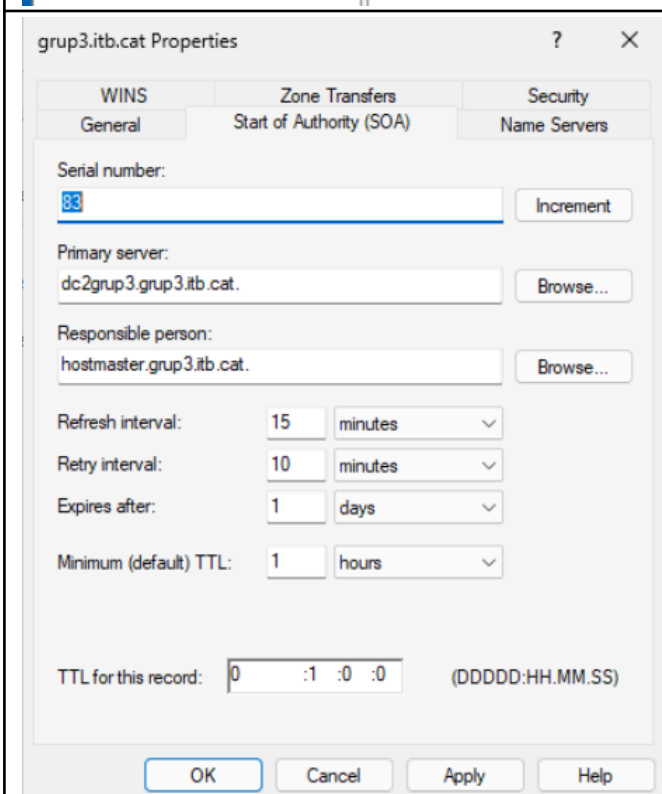
OK Cancel Apply Help

5.

Amb l'eina de Wireshark demostrar que es realitza l'algorisme de DHCP amb tots els paquets DHCP Discover, DHCP Offer, DHCP Request i DHCP ACK (DORA). Indicar i demostrar quina es la mida de cadascun dels paquets DHCP Discover, DHCP Offer, DHCP Request i DHCP ACK



Name	Type	Data	Timestam
_msdcs			
_sites			
_tcp			
_udp			
DomainDnsZones			
ForestDnsZones			
(same as parent folder)	Start of Authority (SOA)	[82], dc2grup3.grup3.itb.c...	static
(same as parent folder)	Name Server (NS)	dc1grup3.grup3.itb.cat.	static
(same as parent folder)	Name Server (NS)	dc2grup3.grup3.itb.cat.	static
(same as parent folder)	Host (A)	192.168.103.1	08/03/202
dc1grup3	Host (A)	192.168.103.1	static
dc2grup3	Host (A)	192.168.120.44	static
dc2grup3	Host (A)	192.168.103.101	static



grup3.itb.cat Properties

WINS Zone Transfers Security

General Start of Authority (SOA) Name Servers

Serial number: 88 Increment

Primary server: dc2grup3.grup3.itb.cat. Browse...

Responsible person: hostmaster.grup3.itb.cat. Browse...

Refresh interval: 15 minutes

Retry interval: 10 minutes

Expires after: 1 days

Minimum (default) TTL: 1 hours

TTL for this record: 0 :1 :0 :0 (DDDD:HH.MM.SS)

OK Cancel Apply Help

Se ha comprobado en el DC2 que, tras unir el cliente DC2grup3 al dominio, el registro de tipo Host (A) se ha creado automáticamente en nuestra zona DNS. Esto demuestra que las actualizaciones dinámicas seguras configuradas en el ejercicio anterior funcionan correctamente y que el DC2 ha replicado la información desde el DC1 de forma inmediata.

Al verificar las propiedades de la zona en el DC2, se observa que el número de serie (Serial number) del registro SOA ha aumentado (de X a Y). Este incremento confirma que el servidor DNS ha detectado cambios en la zona y ha actualizado su base de datos para mantener la consistencia en el dominio grup3.itb.cat.

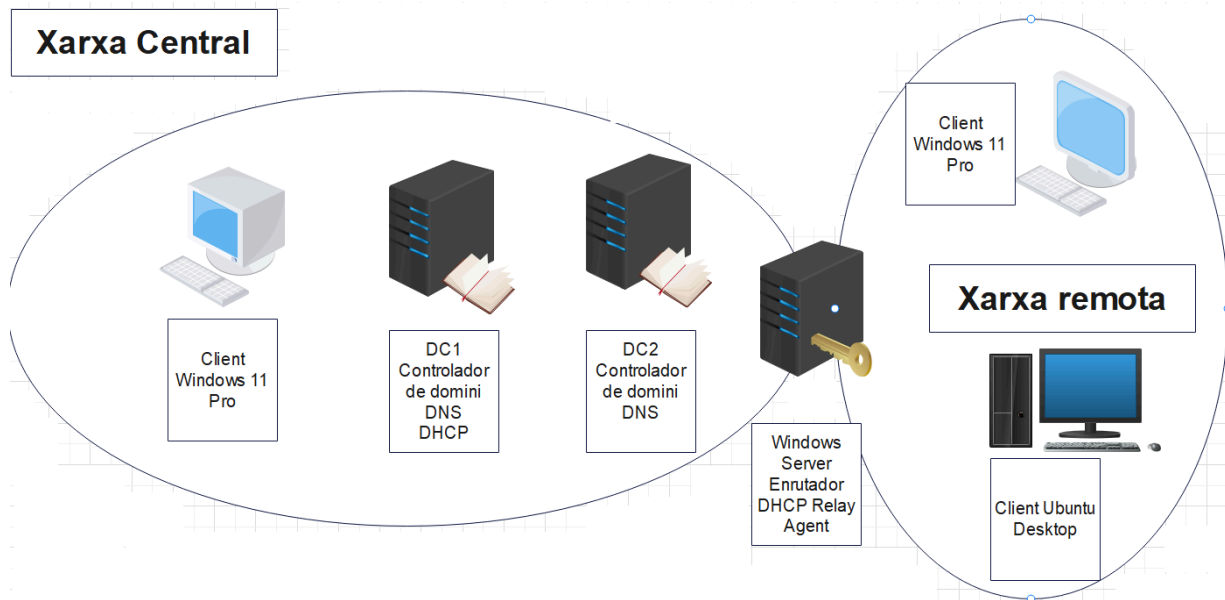
[OBJ]

PUNT DE CONTROL

- Bloc DNS correcte (2 p)
- Bloc DHCP correcte (2 p)

Xarxa Remota Windows Server (Router i DHCP Relay Agent) (6 punts)

Aquesta empresa s'expandeix i obri una delegació amb més de 50 empleats a Madrid.



Per la simulació del router de connexió entre la seu central i la delegació remota farem servir un Windows Server amb la funció d'enrutador.

També volem tenir un únic servidor DHCP central per facilitar l'administració del servei DHCP. Per això crearem un DHCP Relay Agent en aquest servidor perquè s'enrutin tots els paquets de DHCP cap al servidor central.

Instal·lació i configuració del rol d'enrutament en aquest Windows Server.

6.

Webgrafia:

Configurar enrutament en Windows Server

<https://www.youtube.com/watch?v=tAmpjBD1xY>

Verificar que des del client Windows 11 Pro en la xarxa remota tenim connectivitat IP i resolem el nom de domini per DNS.

```
C:\Users\Administrador>ipconfig

Windows IP Configuration

Ethernet adapter Instancia de Ethernet 0:

    Connection-specific DNS Suffix  . : 
    IPv4 Address. . . . . : 192.168.122.116
    Subnet Mask . . . . . : 255.255.252.0
    Default Gateway . . . . . : 192.168.120.1

Ethernet adapter Instancia de Ethernet 0 2:

    Connection-specific DNS Suffix  . : 
    IPv4 Address. . . . . : 192.168.3.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 

Ethernet adapter Instancia de Ethernet 0 3:

    Connection-specific DNS Suffix  . : 
    IPv4 Address. . . . . : 172.16.3.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . :
```

Instal·lació i configuració del rol d'enrutament en aquest Windows Server.

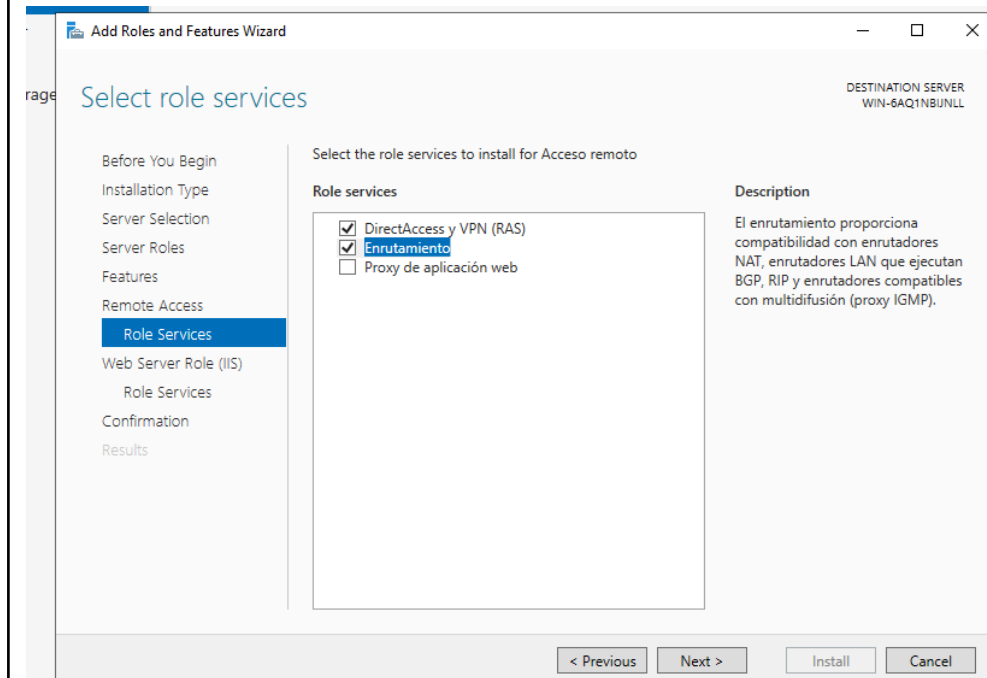
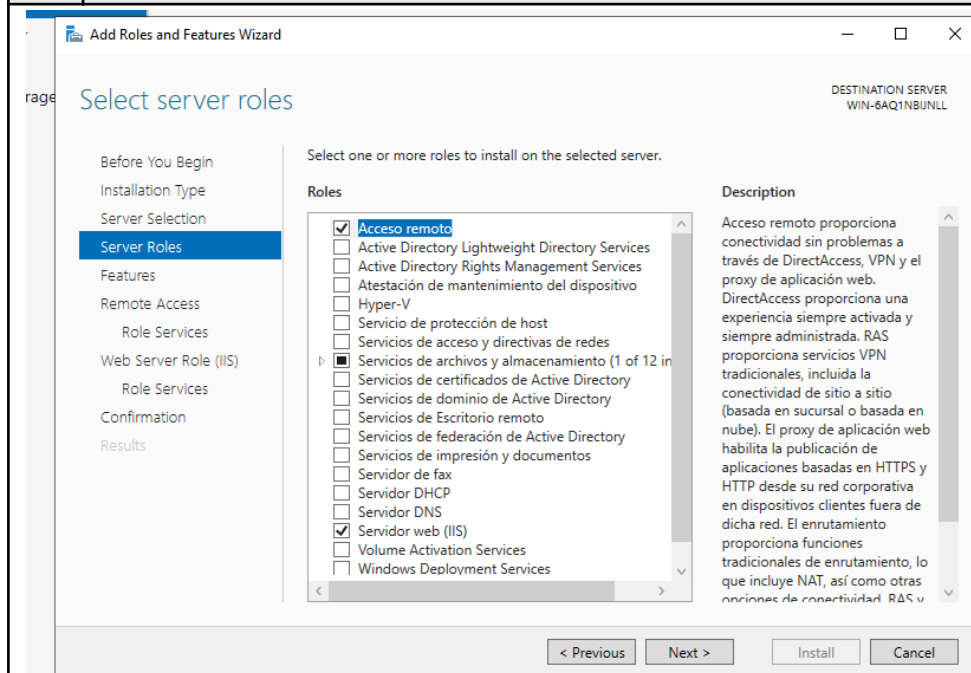
6.

Webgrafia:

Configurar enrutament en Windows Server

<https://www.youtube.com/watch?v=tAmpjBD1xY>

Verificar que des del client Windows 11 Pro en la xarxa remota tenim connectivitat IP i resollem el nom de domini per DNS.



Instal·lació i configuració del rol d'enrutament en aquest Windows Server.

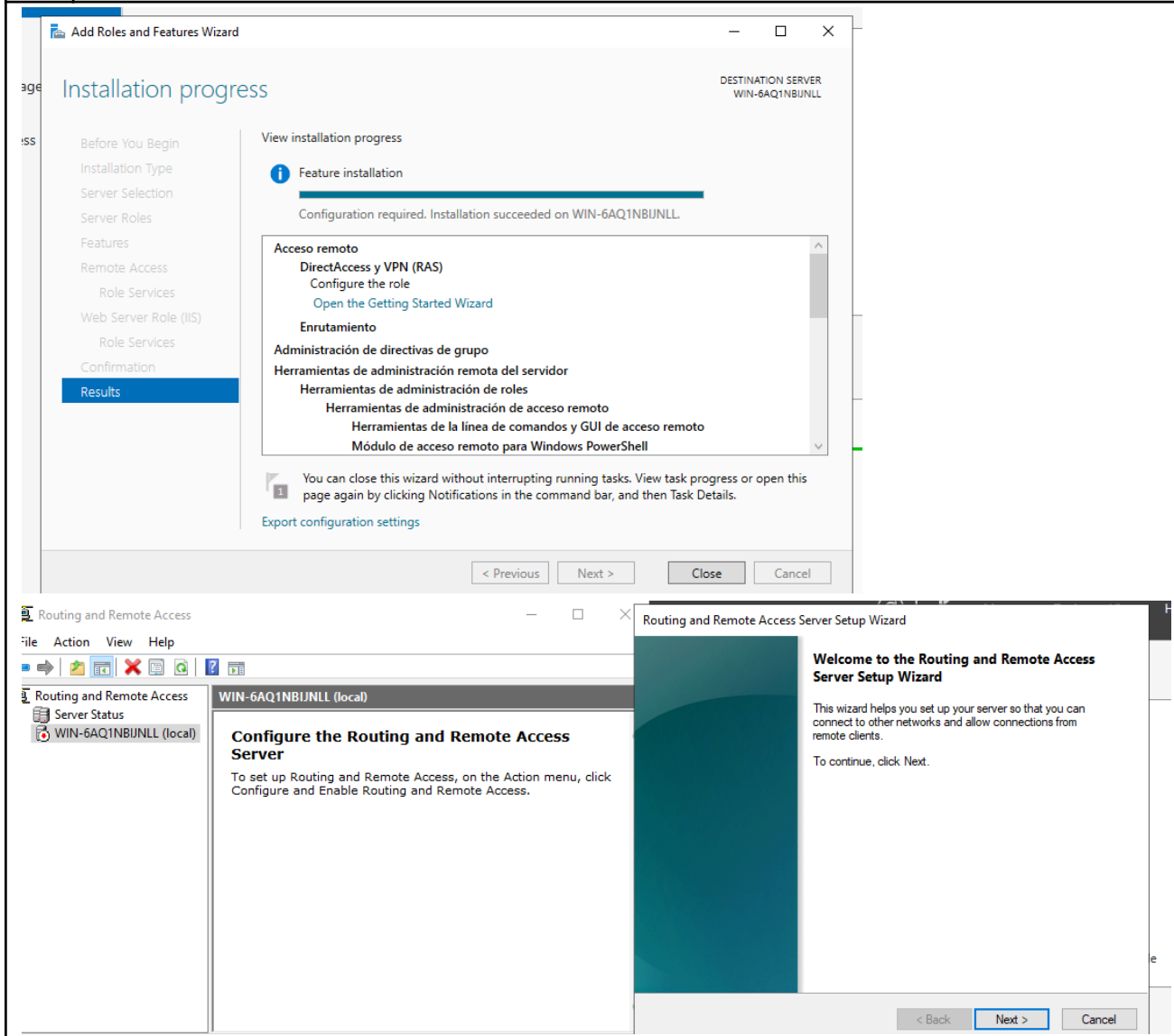
6.

Webgrafia:

Configurar enrutament en Windows Server

<https://www.youtube.com/watch?v=tAmpjBD1xY>

Verificar que des del client Windows 11 Pro en la xarxa remota tenim connectivitat IP i resollem el nom de domini per DNS.



The image displays two screenshots from a Windows Server environment. The top screenshot shows the 'Add Roles and Features Wizard' in the 'Installation progress' stage. The 'Results' section on the left lists the installed features: 'Acceso remoto' (Remote Access), 'Enrutamiento' (Routing), and 'Administración de directivas de grupo' (Group Policy Management). The 'View installation progress' section shows a progress bar for 'Feature installation' and a message: 'Configuration required. Installation succeeded on WIN-6AQ1NBUNLL'. The bottom screenshot shows the 'Routing and Remote Access' console. The 'Server Status' pane on the left shows 'WIN-6AQ1NBUNLL (local)' with a green status icon. The main pane displays the 'Configure the Routing and Remote Access Server' wizard, which includes a 'Welcome to the Routing and Remote Access Server Setup Wizard' message and instructions to click 'Next' to continue.

Instal·lació i configuració del rol d'enrutament en aquest Windows Server.

6.

Webgrafia:

Configurar enrutament en Windows Server

<https://www.youtube.com/watch?v=-tAmpjBD1xY>

Verificar que des del client Windows 11 Pro en la xarxa remota tenim connectivitat IP i resollem el nom de domini per DNS.

Routing and Remote Access Server Setup Wizard

Configuration

You can enable any of the following combinations of services, or you can customize this server.

- ☐ Remote access (dial-up or VPN)
Allow remote clients to connect to this server through either a dial-up connection or a secure virtual private network (VPN) Internet connection.
- ☐ Network address translation (NAT)
Allow internal clients to connect to the Internet using one public IP address.
- ☐ Virtual private network (VPN) access and NAT
Allow remote clients to connect to this server through the Internet and local clients to connect to the Internet using a single public IP address.
- ☐ Secure connection between two private networks
Connect this network to a remote network, such as a branch office.
- ☒ Custom configuration
Select any combination of the features available in Routing and Remote Access.

< Back

Next >

Cancel

Instal·lació i configuració del rol d'enrutament en aquest Windows Server.

6.

Webgrafia:

Configurar enrutament en Windows Server

<https://www.youtube.com/watch?v=-tAmpjBD1xY>

Verificar que des del client Windows 11 Pro en la xarxa remota tenim connectivitat IP i resolem el nom de domini per DNS.

Routing and Remote Access Server Setup Wizard

Custom Configuration

When this wizard closes, you can configure the selected services in the Routing and Remote Access console.

Select the services that you want to enable on this server.

- ☐ VPN access
- ☐ Dial-up access
- ☐ Demand-dial connections (used for branch office routing)
- ☐ NAT
- ☒ LAN routing

< Back Next > Cancel

Routing and Remote Access Server Setup Wizard

Completing the Routing and Remote Access Server Setup Wizard

You have successfully completed the Routing and Remote Access Server Setup wizard.

Summary of selections:

LAN routing

After you close this wizard, configure the selected services in the Routing and Remote Access console.

To close this wizard, click Finish.

< Back Finish Cancel

Routing and Remote Access

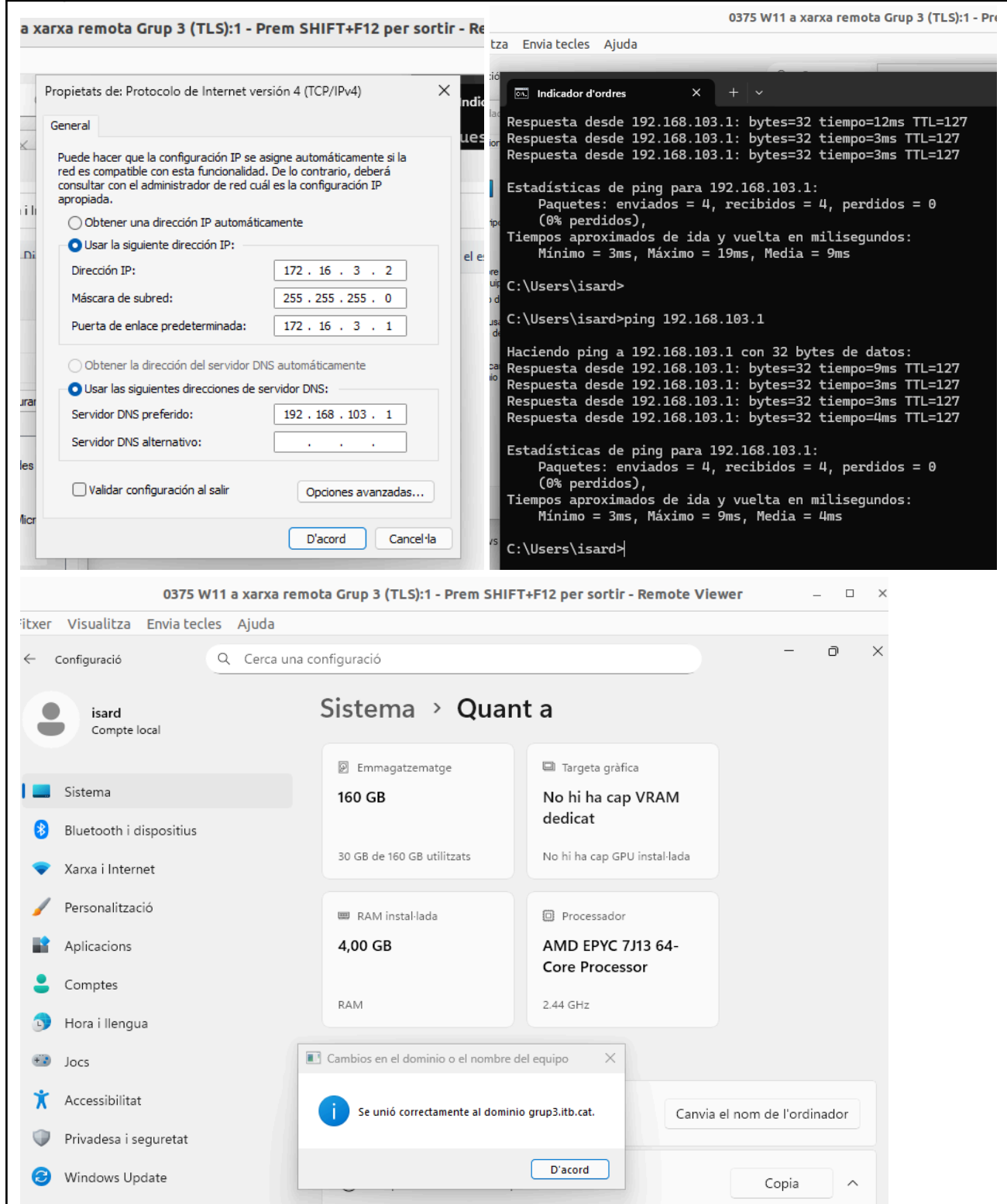
Start the service

The Routing and Remote Access service is ready to use.

Start service Cancel

7.

Una vegada tenim resolució a nivell de domini, unir aquest equip al domini grupX.itb.cat
Verificar que s'afegeix a la zona DNS del DC1 i DC2 aquest nou registre i s'ha augmentat en el SOA el número de sèrie.
Verificar que des del client Ubuntu Desktop en la xarxa remota tenim connectivitat IP i resollem el nom de domini per DNS.



The screenshot shows a Windows environment with two main windows open:

- Propietats de: Protocolo de Internet versión 4 (TCP/IPv4)**: This window shows the network configuration for the '0375 W11 a xarxa remota Grup 3 (TLS):1' connection. The 'General' tab is selected, showing the following settings:
 - Obtener una dirección IP automáticamente: ☐
 - Usar la siguiente dirección IP: ☒
 - Dirección IP: 172.16.3.2
 - Máscara de subred: 255.255.255.0
 - Puerta de enlace predeterminada: 172.16.3.1
 - Obtener la dirección del servidor DNS automáticamente: ☐
 - Usar las siguientes direcciones de servidor DNS: ☒
 - Servidor DNS preferido: 192.168.103.1
 - Servidor DNS alternativo: . . .
 - Validar configuración al salir: ☐
 - Opciones avanzadas... button
 - D'acord and Cancel·la buttons
- 0375 W11 a xarxa remota Grup 3 (TLS):1 - Remote Viewer**: This window shows the system information for the 'isard' user. The 'Sistema > Quant a' section displays:
 - Emmagatzematge: 160 GB (30 GB de 160 GB utilitzats)
 - Tarjeta gráfica: No hi ha cap VRAM dedicat (No hi ha cap GPU instal·lada)
 - RAM instal·lada: 4,00 GB
 - Processador: AMD EPYC 7J13 64-Core Processor (2.44 GHz)

A notification window titled 'Cambios en el dominio o el nombre del equipo' is also visible, stating: 'Se unió correctamente al dominio grup3.itb.cat.' with a 'D'acord' button.

0375 Ubuntu Desktop a xarxa remota Grup 3 (TLS):1 - Prem SHIFT+F12 per sortir - Remote Viewer

via tecles Ajuda

9 de març 11:38

Paràmetres Xarxa

Cablejat

Cancel·la Per cable Aplica

Detalls Identitat IPv4 IPv6 Seguretat

Mètode IPv4

☐ Automàtic (DHCP) ☐ Només enllaç local

☒ Manual ☐ Inhabilita

☐ Compartit a altres ordinadors

Adreces

Adreça	Màscara de xarxa	Passarella	
172.16.3.3	255.255.255.0	172.16.3.1	
			

DNS Automàtic ☒

192.168.103.1

Separa les adreces IP amb comes

0375 Ubuntu Desktop a xarxa remota Grup 3 (TLS):1 - Prem SHIFT+F12 per sortir - Remote Viewer

Visualitza Envia tecles Ajuda

9 de març 11:39

isard@ubuntu-24: ~

```

isard@ubuntu-24:~$ ping 192.168.103.1
PING 192.168.103.1 (192.168.103.1) 56(84) bytes of data.
64 bytes from 192.168.103.1: icmp_seq=1 ttl=127 time=5.05 ms
64 bytes from 192.168.103.1: icmp_seq=2 ttl=127 time=4.99 ms
64 bytes from 192.168.103.1: icmp_seq=3 ttl=127 time=4.81 ms
^C
--- 192.168.103.1 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2003ms
rtt min/avg/max/mdev = 4.814/4.951/5.051/0.100 ms
isard@ubuntu-24:~$ ping 172.16.3.1
PING 172.16.3.1 (172.16.3.1) 56(84) bytes of data.
64 bytes from 172.16.3.1: icmp_seq=1 ttl=128 time=2.89 ms
64 bytes from 172.16.3.1: icmp_seq=2 ttl=128 time=2.46 ms
64 bytes from 172.16.3.1: icmp_seq=3 ttl=128 time=3.02 ms
^C
--- 172.16.3.1 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2004ms
rtt min/avg/max/mdev = 2.456/2.789/3.018/0.241 ms
isard@ubuntu-24:~$

```

grup3.itb.cat Properties

WINS Zone Transfers Security
General Start of Authority (SOA) Name Servers

Serial number:
 Increment

Primary server:
 Browse...

Responsible person:
 Browse...

Refresh interval: minutes

Retry interval: minutes

Expires after: days

Minimum (default) TTL: hours

TTL for this record: : : : (DDDD:HH.MM.SS)

OK Cancel Apply Help

DNS Manager				
File Action View Help				
DNS				
DC2GRUP3 Forward Lookup Zones _msdcs.grup3.itb.cat dc domains gc pdcc grup3.itb.cat Reverse Lookup Zones Trust Points Conditional Forwarders				
Name	Type	Data	Timestamp	
_msdcs				
_sites				
_tcp				
_udp				
DomainDnsZones				
ForestDnsZones				
(same as parent folder)	Start of Authority (SOA)	[87], dc2grup3.grup3.itb.c...	static	
(same as parent folder)	Name Server (NS)	dc2grup3.grup3.itb.cat.	static	
(same as parent folder)	Name Server (NS)	dc1grup3.grup3.itb.cat.	static	
(same as parent folder)	Host (A)	192.168.103.1	08/03/202	
dc1grup3	Host (A)	192.168.103.1	static	
dc2grup3	Host (A)	192.168.103.101	static	
dc2grup3	Host (A)	192.168.120.43	static	
WdCI	Host (A)	172.16.3.10	09/03/202	

Instal·lació i configuració del DHCP Relay Agent.

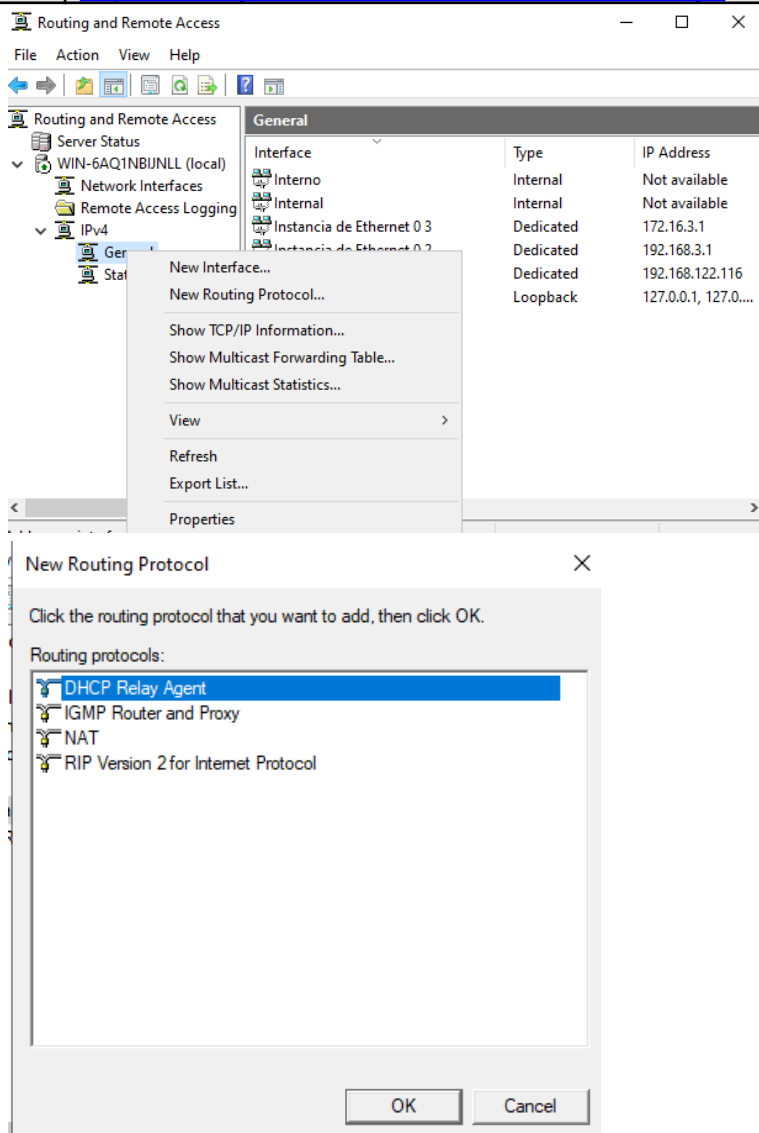
Webgrafia:

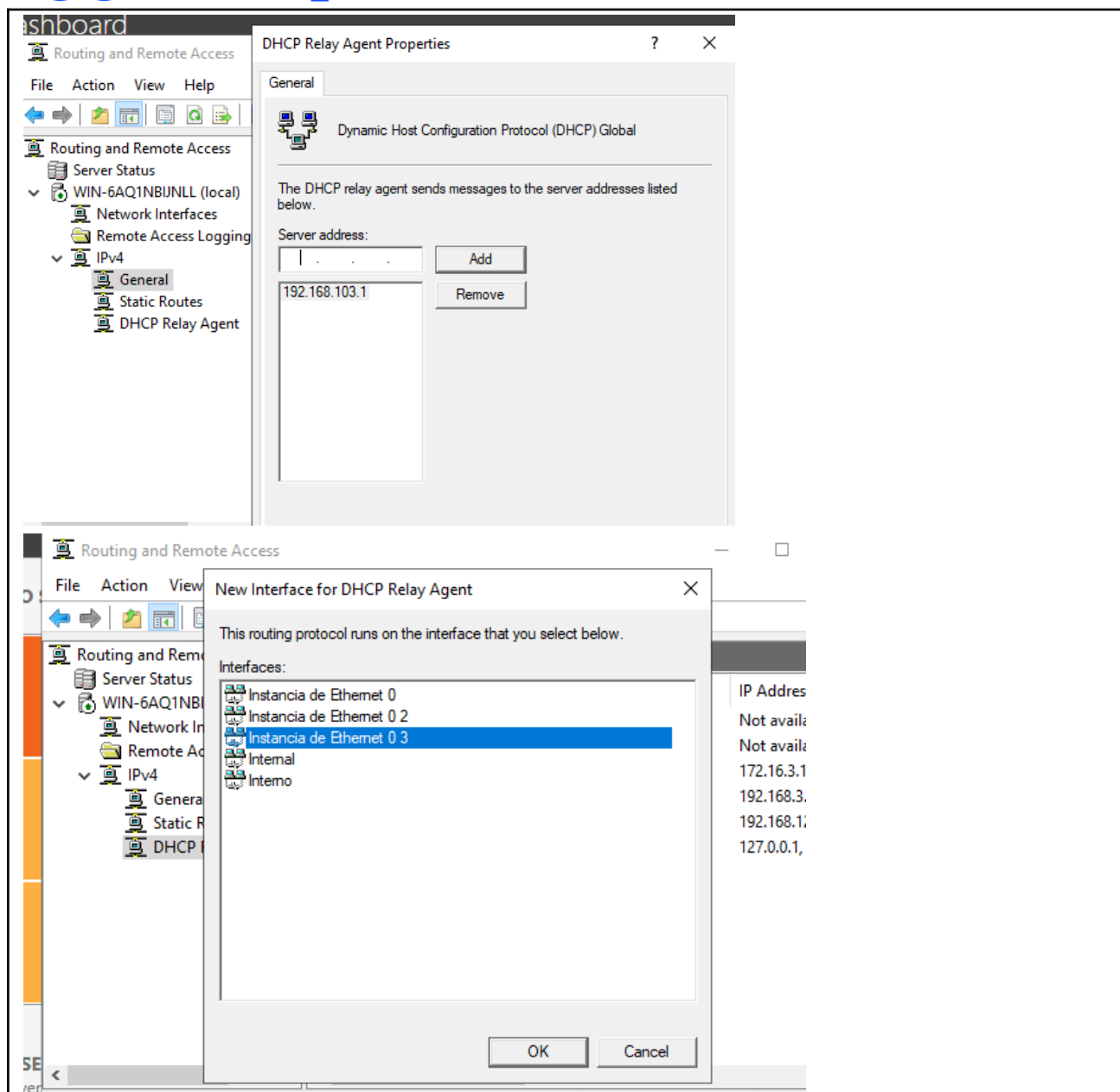
Install a DHCP Relay Agent

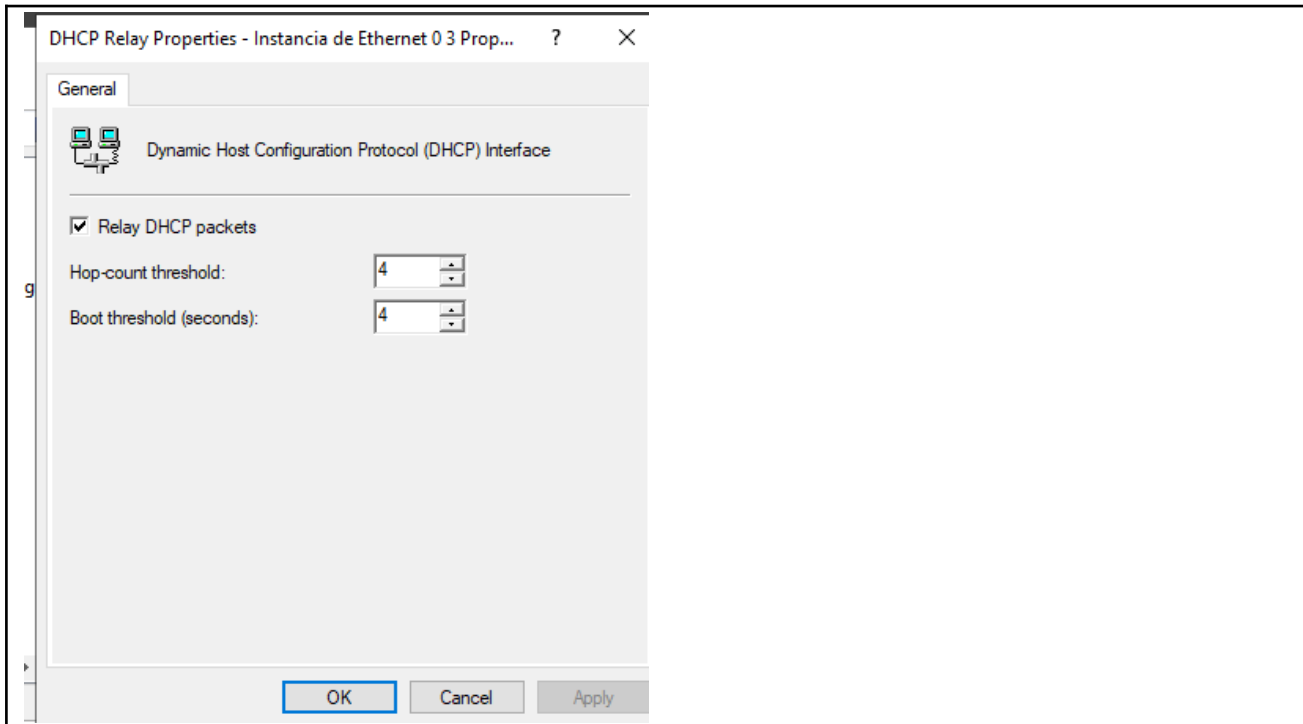
8. <https://learn.microsoft.com/en-us/windows-server/networking/technologies/dhcp/dhcp-deploy-relay-agent>

Configuració de DHCP Relay Agent

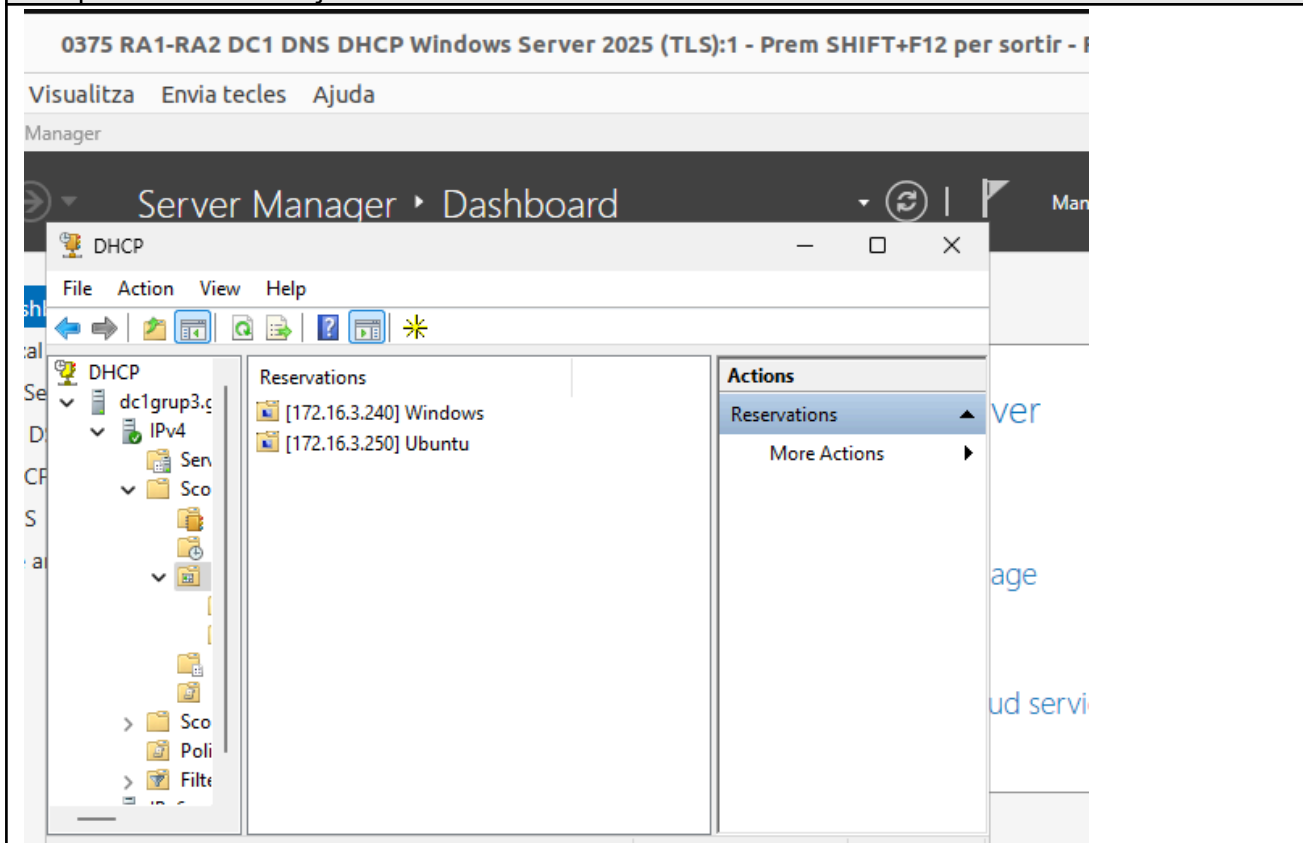
<https://www.youtube.com/watch?v=8US717b3oy0>





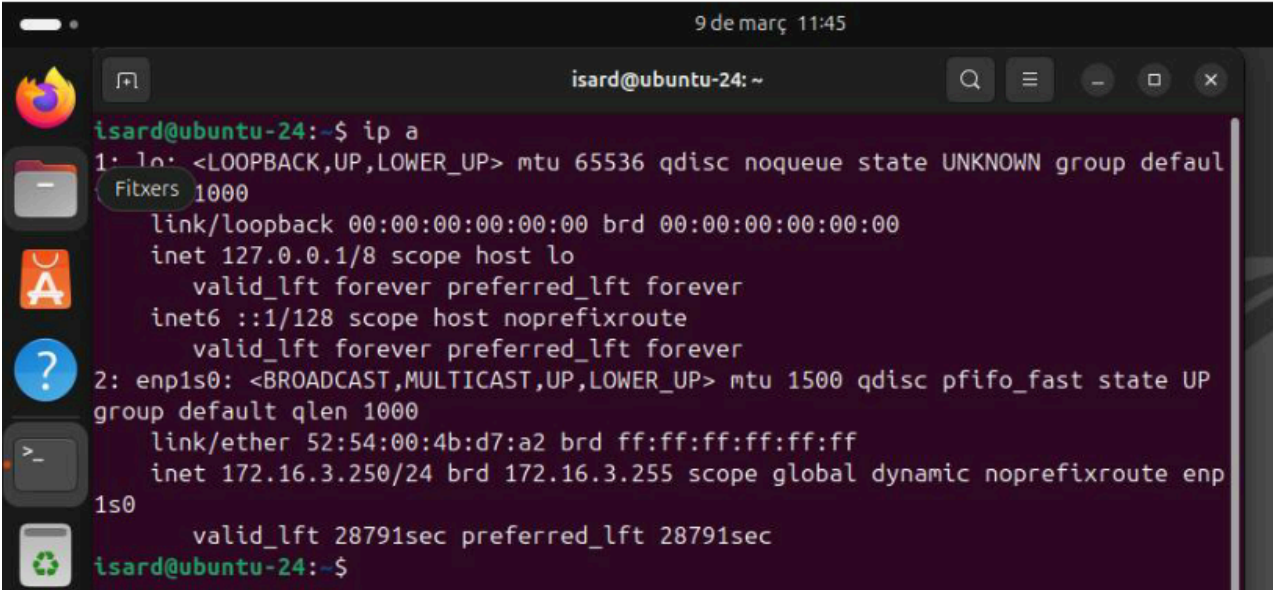


9. Fer que tant el client Windows 11 Pro unit al domini com l'Ubuntu Desktop agafin la configuració automàtica per DHCP.
 Configurar que el client Windows 11 Pro de la xarxa remota faci una reserva per DHCP via MACAddress forçant la IP 172.16.X.240
 Configurar que el client Ubuntu Desktop de la xarxa remota faci una reserva per DHCP via MACAddress forçant la IP 172.16.X.250



0375 Ubuntu Desktop a xarxa remota Grup 3 (TLS):1 - Prem SHIFT+F12 per sortir - Remote Viewer

Fitxer Visualitza Envia tecles Ajuda



```
isard@ubuntu-24:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp1s0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP
    group default qlen 1000
    link/ether 52:54:00:4b:d7:a2 brd ff:ff:ff:ff:ff:ff
    inet 172.16.3.250/24 brd 172.16.3.255 scope global dynamic noprefixroute enp1s0
        valid_lft 28791sec preferred_lft 28791sec
isard@ubuntu-24:~$
```

Adaptador de Ethernet Ethernet 2:

```
Sufijo DNS específico para la conexión. . : grup3.itb.cat
Dirección IPv4. . . . . : 172.16.3.240
Máscara de subred . . . . . : 255.255.255.0
Puerta de enlace predeterminada . . . . : 172.16.3.1
```

C:\Users\isard>

PUNT DE CONTROL

- Bloc Windows Server com enrutador (3 p)
- Bloc Windows Server com DHCP Relay Agent (2 p)
- Clients Windows i Ubuntu remots amb DHCP automàtic i per reserva MAC Address (1 p)